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# LoRaWAN

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## End Device Certification

### Relay Mechanism Test Specification

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Version 1.0.0

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## 47 Revision History

| Version | Author          | Updates         | Date         |
|---------|-----------------|-----------------|--------------|
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|         |                 |                 |              |
|         |                 |                 |              |

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## 50 Glossary

|                           |                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LoRaWAN</b>            | Protocol specification developed and maintained by the LoRa Alliance.                                                                           |
| <b>Certification Logo</b> | LoRa Alliance defined logo that can be displayed on the Certified product and any documentation and marketing information about the End-Device. |
| <b>LoRa Test House</b>    | Organization and corresponding facility accredited by the LoRa Alliance to perform Certification testing.                                       |
| <b>End Device</b>         | Device submitted to a LoRa-Alliance Authorized Test House (ATH) for Certification.                                                              |

## 51 Abbreviations & Acronyms

|             |                                                                                      |
|-------------|--------------------------------------------------------------------------------------|
| <b>ABP</b>  | Activation by Personalization                                                        |
| <b>ADR</b>  | Adaptive Data Rate                                                                   |
| <b>AS</b>   | Application Server                                                                   |
| <b>ATH</b>  | Authorized Test House                                                                |
| <b>AWG</b>  | Arbitrary Waveform Generator                                                         |
| <b>DR</b>   | Data Rate                                                                            |
| <b>DUT</b>  | Device Under Test                                                                    |
| <b>ERP</b>  | Equivalent Radiated Power compared to a dipole antenna (expressed in dBd)            |
| <b>EIRP</b> | Equivalent Isotropic Radiated Power: $ERP = EIRP - 2.15\text{dB}$ (expressed in dBi) |
| <b>ETSI</b> | European Telecommunications Standards Institute                                      |
| <b>FSK</b>  | Frequency Shift Keying modulation technique.                                         |
| <b>GW</b>   | Gateway                                                                              |

|             |                               |
|-------------|-------------------------------|
| <b>LCTT</b> | LoRaWAN Conformance Test Tool |
| <b>MAC</b>  | Media Access Control          |
| <b>NS</b>   | Network Server                |
| <b>OTAA</b> | Over-the-Air Activation       |
| <b>TCL</b>  | Test Control Layer            |
| <b>TRP</b>  | Total Radiated Power          |

## 52 Definitions of terms used in this document

|                    |                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>XXXX</b>        | X is a valid number which would vary based on the region being tested                                                            |
| <b>CP-CMD</b>      | Certification Protocol Command                                                                                                   |
| <b>DC</b>          | Dynamic Channel                                                                                                                  |
| <b>FC</b>          | Fixed Channel                                                                                                                    |
| <b>MAC-CMD</b>     | Medium Access Control Command                                                                                                    |
| <b>MinDR</b>       | Minimum Data Rate for the region tested, as specified in the Regional Parameters Specification RP2 1.0.1                         |
| <b>Max125kHzDR</b> | Maximum Data Rate using 125 kHz bandwidth for the region tested, as specified in the Regional Parameters Specification RP2 1.0.1 |
| <b>R</b>           | Repeat until the condition specified within brackets [] is completed                                                             |

## 53 References

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- [1] LoRaWAN Specification TS001 L2 v1.0.4.
- [2] LoRaWAN Regional Parameters Specification RP002 v1.0.4.
- [3] LoRaWAN Certification Protocol Specification TS009 v1.2.0.
- 55 [4] LoRaWAN End Device Certification Requirements for All Regions v1.6



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- 56 [5] TS011 LoRaWAN Relay Specification v1.0.1
- 57 [6] TS008 FPort Assignments
- 58 [7] Errata-on-the-TS011-1.0.0-Relay-Specification-v1.0
- 59
- 60

## 1. Introduction

This document specifies the testing requirements for an End-Device with support for the Relay Mechanism [5] and [7]. The Device under Test (DUT) may be an End-Device which supports Relay mechanism or a Relay device itself. If the DUT is an End-Device, Section 2 tests must be executed. If the DUT is a Relay device, Section 3 tests must be executed.

Additionally, the End-Device must be tested as a Class A/B/C End-Device as supported according to the LoRaWAN End Device Certification Requirements for All Regions v1.6 [4].

## Test Condition for all tests

The TCL must handle the LoRaWAN protocol like a real network server, unless specified otherwise in the tests.

Throughout the testing, the End-Device MAY send the MAC commands and uplink frames and the TCL must respond appropriately as specified below:

- If the End-Device sends a Confirmed frame anytime, the TCL must respond with an Acknowledgement.
- If the End-Device sends a DeviceTimeReq MAC command or any other End-Device triggered MAC command anytime, the TCL must respond with a DeviceTimeAns MAC answer or the appropriate MAC answer immediately.

## Test Notes

1. **Default Data Rate:** All LinkADRRReq MAC commands in this document must set the Data Rate to Max125kHzDR, unless specified otherwise.
2. **MIC check:** When the tool encounters an invalid MIC, it must fail the specific test being performed.
3. **Missed MAC or Certification Protocol commands:** During testing, if MAC or Certification Protocol commands are missed by the End-Device, the command must be re-sent by the TCL up to two additional times before failing the test.
4. **Missed RxAppCntReq command:** When an RxAppCntReq command is sent from the TCL to the End-Device, if the response is not received from the End-Device, the request must be re-sent up to two additional times.
5. **Downlink after sending MAC commands which require a response:** Unless specified otherwise, for all MAC response commands which require a response from the TCL (namely, RXParamSetupAns, RXTimingSetupAns, DICHannelAns), the TCL must send a downlink immediately after receiving this MAC response command.

6. **Uplink response timing:** The End-Device responses to the TCL downlinks may not necessarily happen at the next uplink from the End-Device. The TCL needs to wait for a End-Device uplink from the appropriate FPort for the application package being tested. If the End-Device sends 3 uplinks for a different FPort, the TCL must resend the downlink and wait up to 3 more uplinks for a response. The downlinks are repeated up to 2 additional times.

## 2. DUT = END-DEVICE

The tests are conducted in a test harness generally comprised of the components represented in Figure 1.

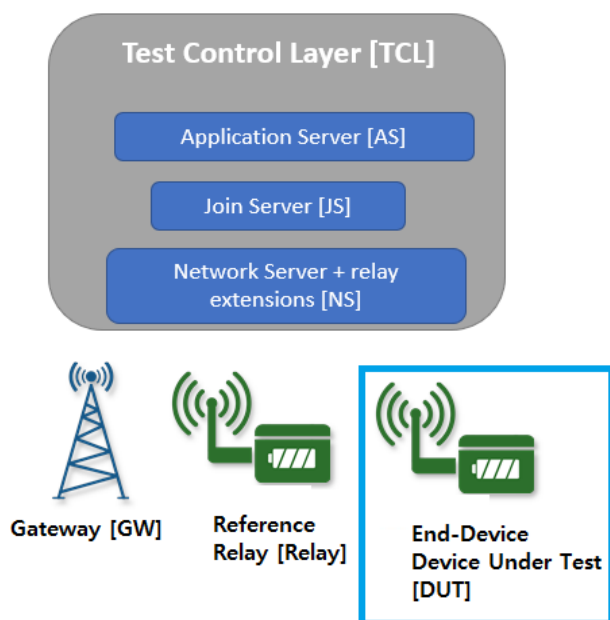


Figure 1 Test Harness Architecture when the DUT = End-Device

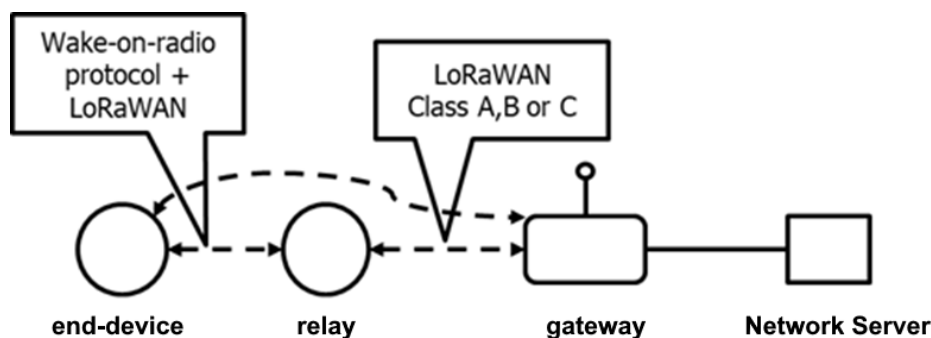


Figure 2 LoRaWAN Relay mechanism principle



The steps to test the End-Device in this document define the commands exchanged between the ED and the TCL or Relay and TCL. The WOR commands exchanged between the ED and Relay are out of scope for the LCTT testing.

## 2.1. Pre-Requisite Setup Test

The Pre-requisite setup tests need to be run as a first step on the Relay and ED before performing any of the other test cases mentioned in this section, to setup the devices and speed up the test execution.

The Pre-requisite tests include-

1. Activation of Relay device (reference Relay)
  - Standard activation (OTAA/ABP)
  - Relay sends an uplink after activation
  - Change TXPeriodicity to 5 seconds
  - Switch to Unconfirmed uplink frames (if not already)
  - Turn on the ADR bit (if not already set)
  - Change the Relay uplink DataRate to Max125KHzDR
  - Enable the Relay with RelayConfReq command
  - Set the relay to no limit with ConfigureFwdLimitReq command
  - Change TXPeriodicity to 120 seconds
2. Activation of DUT (End-Device)
  - 
  - If ED = OTAA device
    - o ED sends Join-Request to TCL
    - o TCL responds with Join-Accept
    - o Change TXPeriodicity to 5 seconds
    - o TCL enables the Relay mode on the ED
    - o ED sends the Join-Request to the Relay
    - o Relay forwards the Join-Request to the TCL
    - o TCL sends UpdateUplinkListReq to the Relay
  - If ED = ABP device
    - o Relay sends NotifyNewEndDeviceReq to TCL
    - o TCL sends UpdateUplinkListReq to the Relay
  - Change TXPeriodicity to 20 seconds
  - Switch to Unconfirmed uplink frames (if not already)
  - Turn on the ADR bit (if not already set)
  - Change the uplink DataRate to Max125KHz



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- 145                    -    TCL sends the EndDeviceConfReq to the ED via the Relay to enable the Relay mode on
- 146                    the ED
- 147                    -    ED responds with the EndDeviceConfAns via the Relay
- 148

## 149    Test Procedure Frame Sequence Chart

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| Step | Procedure                                                                |               |                |                 | Frame Sequence                                                                                                                                    | Remarks                             |
|------|--------------------------------------------------------------------------|---------------|----------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
|      | <b>ACTIVATION of REFERENCE RELAY DEVICE</b>                              | ED<br><><br>R | R<br><><br>TCL | ED<br><><br>TCL |                                                                                                                                                   |                                     |
| 1    | If Relay = OTAA device, Relay sends a Join-Request frame                 |               | →              |                 |                                                                                                                                                   |                                     |
|      | If Relay = ABP device, skip Step 1.<br>TCL sends a Join-Accept frame     |               | ←              |                 |                                                                                                                                                   |                                     |
| 2    | Relay sends Confirmed or Unconfirmed frame                               |               | →              |                 |                                                                                                                                                   |                                     |
|      | The TCL sends Unconfirmed frame to Relay                                 |               | ←              |                 | CP-CMD<br>TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 5<br>Payload = [0x]0601<br><br>If Confirmed uplink, send ACK                     | Change the Periodicity of the Relay |
| 3    | Relay sends Confirmed or Unconfirmed frame                               |               | →              |                 |                                                                                                                                                   |                                     |
|      | If Relay sent Confirmed frame, then TCL sends Unconfirmed frame to Relay |               | ←              |                 | Acknowledge<br><br>CP-CMD TxFramesCtrlReq<br>Frame type = Unconfirmed<br>Payload = [0x]0701                                                       |                                     |
|      | Else, skip this step                                                     |               |                |                 |                                                                                                                                                   |                                     |
| 4    | Relay sends Unconfirmed frame                                            |               | →              |                 |                                                                                                                                                   |                                     |
|      | TCL sends Unconfirmed frame to Relay                                     |               | ←              |                 | If FCtrl ADR Bit = false, then perform this step, else skip to next downlink<br><br>CP-CMD AdrBitChangeReq-ON<br>FPort = 224<br>Payload = [0x]040 |                                     |
| 5    | Relay sends Unconfirmed frame                                            |               | →              |                 |                                                                                                                                                   |                                     |

|                                                                                  |                                             |  |   |  |                                                                                                                                                                                                                                                                                                                                 |                                     |
|----------------------------------------------------------------------------------|---------------------------------------------|--|---|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
|                                                                                  | The TCL sends Unconfirmed frame to Relay    |  | ← |  | MAC-CMD LinkADRReq<br>FPort = 0<br>DataRate = Max125kHzDR, refer [2]<br>Payload = [0x]03XXXXXXXXXX<br>ChMaskCntl:<br>DC = 0,<br>FC = 6<br>ChMask:<br>DC - Enable only default channels<br>FC = [0x]00FF                                                                                                                         |                                     |
| 6                                                                                | Relay sends Unconfirmed frame to TCL        |  | → |  | MAC-CMD LinkADRAns<br>Payload = [0x]0307                                                                                                                                                                                                                                                                                        |                                     |
|                                                                                  | TCL sends Unconfirmed frame to Relay        |  | ← |  | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 0 (1s)<br>DefaultChIdx = first default channel for the region [2]<br>SecondChIdx = 0<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 1 (200kHz)<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX |                                     |
| 7                                                                                | Relay sends an Unconfirmed frame to the TCL |  | → |  | MAC-CMD RelayConfAns<br>Status = 1 for all ACK bits<br>Payload = [0x]40XX                                                                                                                                                                                                                                                       |                                     |
|                                                                                  | TCL sends Unconfirmed frame to Relay        |  | ← |  | MAC-CMD ConfigureFwdLimitReq<br>FPort = 0<br>ResetLimitCounter = 1 (reload rate)<br>JoinReqReloadRate = 127<br>NotifyReloadRate = 127<br>GlobalUplinkReloadRate = 127<br>OverallReloadRate = 127<br>FwdLimitLoadCapacityPL = max size<br>Payload = [0x]45XXXXXXXXXX                                                             |                                     |
| 8                                                                                | Relay sends an Unconfirmed frame to the TCL |  | → |  | MAC-CMD ConfigureFwdLimitAns<br>Payload = [0x]45                                                                                                                                                                                                                                                                                |                                     |
|                                                                                  | The TCL sends Unconfirmed frame to Relay    |  | ← |  | CP-CMD TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 120<br>Payload = [0x]0608                                                                                                                                                                                                                                         | Change the Periodicity of the Relay |
| <b>ACTIVATION of END-DEVICE</b>                                                  |                                             |  |   |  |                                                                                                                                                                                                                                                                                                                                 |                                     |
| If End-Device is an OTAA device, perform this section. Else, skip to ABP section |                                             |  |   |  |                                                                                                                                                                                                                                                                                                                                 |                                     |

|    |                                             |   |   |   |                                                                                                                                                         |                                                  |
|----|---------------------------------------------|---|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 9  | ED must send a Join-Request frame           |   |   | → |                                                                                                                                                         |                                                  |
|    | TCL sends a Join-Accept frame               |   |   | ← |                                                                                                                                                         |                                                  |
| 10 | ED must send Confirmed or Unconfirmed frame |   |   | → |                                                                                                                                                         |                                                  |
|    | TCL sends Unconfirmed frame                 |   |   | ← | If ED sent Confirmed uplink frame, TCL must Acknowledge<br><br>CP-CMD<br>TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 5<br>Payload = [0x]0601 |                                                  |
| 11 | ED must send Confirmed or Unconfirmed frame |   |   | → |                                                                                                                                                         |                                                  |
|    | TCL sends Unconfirmed frame                 |   |   | ← | If ED sent Confirmed uplink frame, TCL must Acknowledge<br><br>CP-CMD RelayModeCtrl<br>Ctrl = 1<br>FPort = 224<br>Payload = [0x]53XX                    |                                                  |
| 12 | ED must send Confirmed or Unconfirmed frame |   |   | → |                                                                                                                                                         |                                                  |
|    | TCL sends Unconfirmed frame                 |   |   | ← | If ED sent Confirmed uplink frame, TCL must Acknowledge<br><br>CP-CMD DutJoinReq<br>FPort = 224<br>Payload = [0x]02                                     |                                                  |
| 13 | ED sends Unconfirmed frame to Relay         | → |   |   | <i>WORRelayJoin and PHYPayload which includes the Join-Request</i>                                                                                      | These messages cannot be verified                |
|    | Relay sends Unconfirmed frame to the TCL    |   | → |   | ForwardUplinkReq (UplinkMetaData, UplinkFrequency, PHYPayload containing Join-Request)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXX               |                                                  |
|    | TCL sends Unconfirmed frame to Relay        |   | ← |   | ForwardDownlinkReq (PHYPayload containing Join-Accept)<br>FPort = 226<br>Payload = [0x]variable                                                         | Relay in turn sends the Join-Accept to ED on RXR |
| 14 | Relay sends Unconfirmed frame to the TCL    |   | → |   | Empty uplink                                                                                                                                            |                                                  |

|                                                                  |                                                        |   |   |   |                                                                                                                                                                                                                     |  |
|------------------------------------------------------------------|--------------------------------------------------------|---|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                                                                  | TCL sends Unconfirmed frame to Relay on RX2 window     |   | ← |   | MAC-CMD<br>UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 3<br>UplinkLimitReloadRate = 63<br>RootWorSKey of ED<br>DevAddr of ED<br>WFCnt32 = 0<br>Payload = [0x]43 XX (26 octets) |  |
| 15                                                               | Relay must send Unconfirmed frame to the TCL           |   | → |   | MAC-CMD<br>UpdateUplinkListAns<br>Payload = [0x]43                                                                                                                                                                  |  |
| 16                                                               | ED sends a Confirmed or Unconfirmed frame              | → |   |   |                                                                                                                                                                                                                     |  |
|                                                                  | The TCL sends Unconfirmed frame to the ED directly     |   |   | ← | CP-CMD<br>TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 20<br>Payload = [0x]0603<br>If ED uplink was a Confirmed frame, send ACK                                                                           |  |
| End of OTAA activation section                                   |                                                        |   |   |   |                                                                                                                                                                                                                     |  |
| <b>If ED is an ABP device, perform the tests in this section</b> |                                                        |   |   |   |                                                                                                                                                                                                                     |  |
| 17                                                               | ED sends Confirmed or Unconfirmed frame to Relay       |   |   | → | <i>WOR RelayUplink</i>                                                                                                                                                                                              |  |
|                                                                  | TCL sends Unconfirmed frame to ED on RX1 window        |   |   | ← | CP-CMD<br>TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 20<br>Payload = [0x]0603<br><br>If ED uplink was a Confirmed frame, send ACK                                                                       |  |
| 18                                                               | Relay sends Unconfirmed frame to the TCL               |   | → |   | MAC-CMD<br>NotifyNewEndDeviceReq<br>DevAddr of ED<br>Payload = [0x]46XXXXXXXXXXXXXX                                                                                                                                 |  |
|                                                                  | TCL sends Unconfirmed frame to Relay                   |   | ← |   | MAC-CMD<br>UpdateUplinkListReq<br>FPort = 0<br>RootWorSKey of ED<br>WFCnt32 = next expected value<br>UplinkLimitBucketSize = 2<br>UplinkLimitReloadRate = 63<br>DevAddr<br>Payload = [0x]43 XX (26 octets)          |  |
| 19                                                               | Relay must send Unconfirmed frame to the TCL           |   | → |   | MAC-CMD<br>UpdateUplinkListAns<br>Payload = [0x]43                                                                                                                                                                  |  |
| End of ABP activation section                                    |                                                        |   |   |   |                                                                                                                                                                                                                     |  |
| 20                                                               | ED sends a Confirmed or Unconfirmed frame to the Relay | → |   |   | <i>PHYPayload</i>                                                                                                                                                                                                   |  |

|    |                                              |   |   |  |                                                                                                                                                                                                                                                                                 |  |
|----|----------------------------------------------|---|---|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    | Relay sends Unconfirmed frame to TCL         |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                    |  |
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | PHYPayload includes CP-CMD TxFramesCtrlReq<br>FPort = 224<br>Frame type = Unconfirmed<br>Payload = [0x]0701<br>If ED uplink was a Confirmed frame, TCL must acknowledge and perform this step, else skip to next downlink<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226 |  |
| 21 | ED sends an Unconfirmed frame to the Relay   | → |   |  | PHYPayload                                                                                                                                                                                                                                                                      |  |
|    | Relay sends Unconfirmed frame to TCL         |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                    |  |
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | If FCtrl ADR Bit = false, then perform this step, else skip to next downlink<br><br>PHYPayload includes CP-CMD AdrBitChangeReq-ON<br>FPort = 224<br>Payload = [0x]040<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226                                                     |  |
| 22 | ED sends an Unconfirmed frame to the Relay   | → |   |  | PHYPayload                                                                                                                                                                                                                                                                      |  |
|    | Relay sends Unconfirmed frame to TCL         |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                    |  |

|    |                                              |   |   |  |                                                                                                                                                                                                                                                                                                                                                                                                                          |              |
|----|----------------------------------------------|---|---|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | <p>PHYPayload includes<br/>MAC-CMD LinkADRReq<br/>DataRate = Max125kHzDR,<br/>refer [2]<br/>Payload = [0x]03XXXXXXXXXX<br/>ChMaskCntl:<br/>    DC = 0,<br/>    FC = 6<br/>ChMask:<br/>    DC - Enable only default<br/>channels<br/>FC = [0x]00FF</p> <p>ForwardDownlinkReq<br/>(PHYPayload)<br/>FPort = 226</p>                                                                                                         |              |
| 23 | ED sends an Unconfirmed frame to the Relay   | → |   |  | <p>PHYPayload includes<br/>MAC-CMD LinkADRAns<br/>Payload = [0x]0307</p>                                                                                                                                                                                                                                                                                                                                                 |              |
|    | Relay sends Unconfirmed frame to TCL         |   | → |  | <p>PHYPayload includes<br/>MAC-CMD LinkADRAns<br/>Payload = [0x]0307</p> <p>ForwardUplinkReq<br/>(UplinkMetadata, Uplink<br/>Frequency, PHYPayload)<br/>FPort = 226<br/>Payload =<br/>[0x]XXXXXXXXXXXXXX</p>                                                                                                                                                                                                             |              |
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | <p>PHYPayload includes<br/>MAC-CMD<br/>EndDeviceConfReq<br/>FPort = 0<br/>RelayModeActivation = 1<br/>(enable)<br/>Back-off = 0<br/>SecondChIdx = 0<br/>SecondChDr = Relay's second<br/>channel DR<br/>SecondChAckOffset = Relay's<br/>second channel ACKOffset<br/>SecondChFreq = Relay's<br/>second channel freq<br/>Payload =<br/>[0x]41XXXXXXXXXXXXXX</p> <p>ForwardDownlinkReq<br/>(PHYPayload)<br/>FPort = 226</p> |              |
| 24 | ED sends an Unconfirmed frame to the Relay   | → |   |  | <p>PHYPayload includes<br/>MAC-CMD<br/>EndDeviceConfAns<br/>Payload = [0x]41XF</p>                                                                                                                                                                                                                                                                                                                                       | Enable Relay |



|  |                                            |  |   |  |                                                                                                                                                                                                       |  |
|--|--------------------------------------------|--|---|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | Relay sends<br>Unconfirmed frame to<br>TCL |  | → |  | PHYPayload includes<br>MAC-CMD<br>EndDeviceConfAns<br>Payload = [0x]41XF<br><br>ForwardUplinkReq<br>(UplinkMetadata, Uplink<br>Frequency, PHYPayload)<br>FPort = 226<br>Payload =<br>[0x]XXXXXXXXXXXX |  |
|--|--------------------------------------------|--|---|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

## 2.2. End-Device MAC command- EndDeviceConf

- Disable the Duty Cycle with CP-CMD RegionalDutyCycleCtrlReq OFF
- TCL sends the MAC command EndDeviceConfReq with RelayModeActivation = 0 (disable) to the End-Device
  - End-Device must reply with EndDeviceConfAns without any error via the Relay.
  - The next uplink must be sent directly to the TCL via the gateway and must not be sent to the Relay.
- TCL sends the MAC command EndDeviceConfReq with
  - RelayModeActivation = 2 (Dynamic) to the End-Device
  - Backoff = 0
  - SmartEnableLevel = 0
  - SecondChIdx = 0
  - SecondChDr = Relay's second channel Data Rate
  - SecondChAckOffset = Relay's second channel ACKoffset
  - SecondChFreq = Relay's second channel frequency
  - End-Device must respond with EndDeviceConfAns with bits 0-3 set to 1
  - End-Device must send the next Uplink to the TCL directly
- TCL does not respond with the Downlink
  - End-Device must send 7 more uplinks to the TCL directly or via the Relay
- TCL does not respond with the Downlinks for all the uplinks
  - End-Device must send a WOR Relay uplink to the Relay for the next uplink initiation instead of sending it directly to the TCL.
- TCL then disables the Relay mode with RelayModeActivation = 0
  - End-Device must send the next uplink directly to the TCL

### Check for Invalid configuration

- TCL sends the MAC command EndDeviceConfReq with
  - RelayModeActivation = 1

- 182                   ▪ Backoff = 0
- 183                   ▪ SmartEnableLevel = 0
- 184                   ▪ SecondChIdx = 1
- 185                   ▪ SecondChDr = any valid second channel DR [2]
- 186                   ▪ SecondChAckOffset = any valid second channel offset value [2]
- 187                   ▪ SecondChFreq = Invalid out of band value [2]
- 188               ○ End-Device must respond with EndDeviceConfAns with Payload NOT = [0x]410F
- 189   • TCL sends the MAC command EndDeviceConfReq with
  - 190                   ▪ RelayModeActivation = 1
  - 191                   ▪ Backoff = 0
  - 192                   ▪ SmartEnableLevel = 0
  - 193                   ▪ SecondChIdx = 1
  - 194                   ▪ SecondChDr = any second channel valid DR [2]
  - 195                   ▪ SecondChAckOffset = Invalid out of band value [2]
  - 196                   ▪ SecondChFreq = any valid second channel freq [2]
- 197               ○ End-Device must respond with EndDeviceConfAns with Payload NOT = [0x]410F
- 198
- 199   • TCL sends the MAC command EndDeviceConfReq with
  - 200                   ▪ RelayModeActivation = 1
  - 201                   ▪ Backoff = 0
  - 202                   ▪ SmartEnableLevel = 0
  - 203                   ▪ SecondChIdx = 1
  - 204                   ▪ SecondChDr = Invalid out of band value [2]
  - 205                   ▪ SecondChAckOffset = any valid second channel offset [2]
  - 206                   ▪ SecondChFreq = any valid second channel freq [2]
- 207               ○ End-Device must respond with EndDeviceConfAns with Payload NOT = [0x]410F
- 208
- 209   • If the Relay belongs to a region that supports LR-FHSS [2], TCL sends the MAC command
  - 210                   EndDeviceConfReq with
    - 211                       ▪ RelayModeActivation = 1
    - 212                       ▪ Backoff = 0
    - 213                       ▪ SmartEnableLevel = 0
    - 214                       ▪ SecondChIdx = 1
    - 215                       ▪ SecondChDr = LR-FHSS DR [2] (invalid)
    - 216                       ▪ SecondChAckOffset = any valid second channel offset [2]
    - 217                       ▪ SecondChFreq = any valid second channel freq [2]
  - 218                   ○ End-Device must respond with EndDeviceConfAns with Payload NOT = [0x]410F
- 219
- 220   • If the Relay belongs to a region that supports FSK [2], TCL sends the MAC command
  - 221                   EndDeviceConfReq with
    - 222                       ▪ RelayModeActivation = 1

- 223                   ▪ Backoff = 0
- 224                   ▪ SmartEnableLevel = 0
- 225                   ▪ SecondChIdx = 1
- 226                   ▪ SecondChDr = FSK [2] (invalid)
- 227                   ▪ SecondChAckOffset = any valid second channel offset [2]
- 228                   ▪ SecondChFreq = any valid second channel freq [2]
- 229           ○ End-Device must respond with EndDeviceConfAns with Payload NOT = [0x]410F

230

- 231   • TCL sends the MAC command EndDeviceConfReq with
  - 232                   ▪ RelayModeActivation = 1
  - 233                   ▪ Backoff = 0
  - 234                   ▪ SmartEnableLevel = 0
  - 235                   ▪ SecondChIdx = 2 (invalid)
  - 236                   ▪ SecondChDr = Relay's second channel DR
  - 237                   ▪ SecondChAckOffset = Relay's second channel offset
  - 238                   ▪ SecondChFreq = Relay's second channel freq
- 239   ○ End-Device must respond with EndDeviceConfAns with Payload NOT = [0x]410F

240

#### 241   Default channel usage

- 242   • TCL sends the MAC command EndDeviceConfReq with
  - 243                   ▪ RelayModeActivation = 1
  - 244                   ▪ Backoff = 0
  - 245                   ▪ SmartEnableLevel = 0
  - 246                   ▪ SecondChIdx = 0
  - 247                   ▪ SecondChDr = Relay's second channel DR
  - 248                   ▪ SecondChAckOffset = Relay's second channel offset
  - 249                   ▪ SecondChFreq = Relay's second channel freq
- 250   ○ End-Device must respond with EndDeviceConfAns with bits 0-3 set to 1 through the
- 251   Relay
- 252   • TCL sends RelayConfReq to the Relay to disable the second channel of the Relay
- 253   ○ Relay responds with a RelayConfAns
- 254   ○ End-Device must send the next uplink to the Relay using the Relay default channel
- 255   settings

256

257

#### 258   **Backoff > 0**

- 259   • TCL disables the Relay mode of the ED by sending the EndDeviceConfReq
- 260   • TCL obtains the WFCntUp of the Relay using the CtrlUplinkListReq
- 261   • TCL sends the MAC command EndDeviceConfReq with Backoff = 5
  - 262                   ▪ RelayModeActivation = 1
  - 263                   ▪ Backoff = 5

- 264
  - SmartEnableLevel = 0
- 265
  - SecondChIdx = 0
- 266
  - SecondChDr = Relay's second channel DR
- 267
  - SecondChAckOffset = Relay's second channel offset
- 268
  - SecondChFreq = Relay's second channel freq
- 269
  - End-Device must NOT respond with EndDeviceConfAns with bits 0-3 set to 1 through
- 270
  - the Relay
- 271
  - TCL sends the Relay a UpdateUplinkListReq with the following parameters
- 272
  - UpdateListIdx = 1
- 273
  - UpdateLimitBucketSize = 3
- 274
  - UpdateLimitReloadRate = 63
- 275
  - DevAddr of the End-Device
- 276
  - WFCnt32 = WFCnt32 + 10
- 277
  - RootWorSKey = valid key
- 278
  - Relay must respond with the UpdateUplinkListAns MAC command
- 279
- 280
- 281
  - ED sends the WOR frames to the Relay and the Relay thus sends a
- 282
  - NotifyNewEndDeviceReq to the TCL continuously
- 283
  - Repeat 5 times
- 284
  - TCL verifies that the ED sends the next uplink directly to the TCL through the gateway only.
- 285
  - ED sends the next uplink via the Relay and Gateway
- 286
- 287
- 288
  - TCL sends the Relay a UpdateUplinkListReq with the valid parameters
- 289
  - UpdateListIdx = 1
- 290
  - UpdateLimitBucketSize = 3
- 291
  - UpdateLimitReloadRate = 63
- 292
  - DevAddr of the End-Device
- 293
  - Valid WFCnt32
- 294
  - RootWorSKey = valid key associated to the DevAddr of the End-Device
- 295
  - Relay must respond with the UpdateUplinkListAns MAC command
- 296
- 297
  - TCL sends the MAC command EndDeviceConfReq with Backoff = 0
- 298
  - RelayModeActivation = 1
- 299
  - Backoff = 0
- 300
  - SmartEnableLevel = 0
- 301
  - SecondChIdx = 0
- 302
  - SecondChDr = Relay's second channel DR
- 303
  - SecondChAckOffset = Relay's second channel offset
- 304
  - SecondChFreq = Relay's second channel freq

- 305                   ○ End-Device must respond with EndDeviceConfAns with bits 0-3 set to 1 through the  
306                   Relay  
307  
308                   • TCL sends the MAC command RelayConfReq to revert to default values. Relay  
309                   responds successfully  
310                   • Enable the Duty Cycle with CP-CMD RegionalDutyCycleCtrlReq ON (default)

311           **Test Procedure Frame Sequence Chart**  
312

| Step | Procedure                                                                                                      | ED<br><><br>R | R<br><><br>TCL | ED<br><><br>TCL | Frame                                                                                                                                                                                                         | Remarks                            |
|------|----------------------------------------------------------------------------------------------------------------|---------------|----------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1    | ED sends an Unconfirmed frame to the Relay                                                                     | →             |                |                 | <i>PHYPayload</i>                                                                                                                                                                                             |                                    |
|      | Relay sends an Unconfirmed frame to the TCL                                                                    |               | →              |                 | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                  |                                    |
|      | The TCL sends Unconfirmed frames                                                                               |               | ←              |                 | PHYPayload includes<br>Confirmed downlink<br>CP-CMD RegionalDutyCycleCtrlReq-<br>OFF<br>FPort = 224<br>Payload = [0x]0500<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX | Turn OFF<br>Regional Duty<br>Cycle |
| 2    | ED sends an Unconfirmed frame to the Relay                                                                     | →             |                |                 | <i>PHYPayload includes<br/>ACK bit = 1</i>                                                                                                                                                                    |                                    |
|      | Relay sends an Unconfirmed frame to the TCL                                                                    |               | →              |                 | PHYPayload includes<br>ACK bit = 1<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                        |                                    |
|      | TCL sends Unconfirmed frame to ED directly                                                                     |               |                | ←               | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 0 (disable)<br>Payload = [0x]41XXXXXXXXXXXX                                                                                                    | Disable Relay                      |
| 3    | ED must send Unconfirmed frame<br><br>Note: ED may send the Unconfirmed frame via Relay or to the TCL directly | →             |                |                 | <i>PHYPayload includes<br/>MAC-CMD EndDeviceConfAns<br/>Payload = [0x]41XF</i>                                                                                                                                |                                    |
|      |                                                                                                                |               |                |                 |                                                                                                                                                                                                               |                                    |
| 4    | ED sends Unconfirmed frame to the TCL directly                                                                 |               |                | →               |                                                                                                                                                                                                               |                                    |

|   |                                                                                                                                                                                                                          |   |   |           |                                                                                                                                                                                                                                                                                                                                                                                                    |                    |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
|   | TCL sends Unconfirmed frame to ED                                                                                                                                                                                        |   | ← |           | PHYPayload includes<br>MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 2 (dynamic)<br>Back-off = 0<br>SmartEnableLevel = 0<br>SecondChIdx = 0<br>SecondChDr = Relay's second channel DR<br>SecondChAckOffset = Relay's second channel ACKoffset<br>SecondChFreq = Relay's second channel freq<br>Payload = [0x]41XXXXXXXXXXXX<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226 | Relay Dynamic mode |
| 5 | ED must send Unconfirmed frame to TCL directly                                                                                                                                                                           |   |   | →         | MAC-CMD EndDeviceConfAns<br>Payload = [0x]41XF                                                                                                                                                                                                                                                                                                                                                     |                    |
| 6 | ED must send Unconfirmed frame<br><br>FCntUp = x<br><br>This frame must be sent 7 times to the TCL<br><br>Note: ED may send the Unconfirmed frame via Relay or to the TCL directly<br><br>TCL does not respond to the ED |   |   | →<br>R[7] |                                                                                                                                                                                                                                                                                                                                                                                                    |                    |
| 7 | ED sends Unconfirmed frame to Relay<br><br>FCntUp = x + 7 (tolerance of 5%)<br><br>Relay sends an Unconfirmed frame to the TCL<br><br>TCL sends Unconfirmed frame to ED directly                                         | → |   |           | PHYPayload<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                                                                                     |                    |
|   |                                                                                                                                                                                                                          |   | → |           | FPort = 0<br><br>MAC-CMD EndDeviceConfReq<br>RelayModeActivation = 0 (disable)<br><br>Payload = [0x]41XXXXXXXXXXXX                                                                                                                                                                                                                                                                                 | Disable Relay      |
| 8 | ED must send Unconfirmed frame to TCL directly                                                                                                                                                                           |   |   | →         | MAC-CMD EndDeviceConfAns<br>Payload = [0x]41XF                                                                                                                                                                                                                                                                                                                                                     |                    |

|    |                                                                                               |  |  |   |                                                                                                                                                                                                                                                                                                                                        |                           |
|----|-----------------------------------------------------------------------------------------------|--|--|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
|    | TCL sends Unconfirmed frame to ED                                                             |  |  | ← | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 1 (enable)<br>Back-off = 0<br>SmartEnableLevel = 0<br>SecondChIdx = 1<br>SecondChDr = Relay's second channel DR<br>SecondChAckOffset = Relay's second channel ACKoffset<br>SecondChFreq = Any invalid second channel freq (invalid) [2]<br>Payload = [0x]41XXXXXXXXXXXX | Invalid SecondChFreq      |
| 9  | ED must send Unconfirmed frame to TCL directly                                                |  |  | → | MAC-CMD EndDeviceConfAns<br>Payload NOT = [0x]410F                                                                                                                                                                                                                                                                                     |                           |
|    | TCL sends Unconfirmed frame to ED                                                             |  |  | ← | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 1 (enable)<br>Back-off = 0<br>SmartEnableLevel = 0<br>SecondChIdx = 1<br>SecondChDr = Relay's second channel DR<br>SecondChAckOffset = Invalid out of band value [2]<br>SecondChFreq = Relay's second channel Freq<br>Payload = [0x]41XXXXXXXXXXXX                      | Invalid SecondChAckOffset |
| 10 | ED must send Unconfirmed frame to TCL directly                                                |  |  | → | MAC-CMD EndDeviceConfAns<br>Payload NOT = [0x]410F                                                                                                                                                                                                                                                                                     |                           |
|    | TCL sends Unconfirmed frame to ED                                                             |  |  | ← | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 1 (enable)<br>Back-off = 0<br>SmartEnableLevel = 0<br>SecondChIdx = 1<br>SecondChDr = Any invalid second channel DR (invalid) [2]<br>SecondChAckOffset = Relay's second channel ACKoffset<br>SecondChFreq = Relay's second channel freq<br>Payload = [0x]41XXXXXXXXXXXX | Invalid SecondChDr        |
| 11 | ED must send Unconfirmed frame to TCL directly                                                |  |  | → | MAC-CMD EndDeviceConfAns<br>Payload NOT = [0x]410F                                                                                                                                                                                                                                                                                     |                           |
|    | If the Relay belongs to a region that supports LR-FHSS [2], TCL sends Unconfirmed frame to ED |  |  | ← | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 1 (enable)<br>Back-off = 0<br>SmartEnableLevel = 0<br>SecondChIdx = 1<br>SecondChDr = LR-FHSS DR (invalid) [2]<br>SecondChAckOffset = Relay's second channel ACKoffset<br>SecondChFreq = Relay's second channel freq<br>Payload = [0x]41XXXXXXXXXXXX                    | SecondChDr = LR-FHSS DR   |



|    |                                                                                                            |   |  |   |                                                                                                                                                                                                                                                                                                                                               |                                          |
|----|------------------------------------------------------------------------------------------------------------|---|--|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 12 | If the Relay belongs to a region that supports LR-FHSS [2], ED must send Unconfirmed frame to TCL directly |   |  | → | MAC-CMD EndDeviceConfAns<br>Payload NOT = [0x]410F                                                                                                                                                                                                                                                                                            |                                          |
|    | If the Relay belongs to a region that supports FSK [2], TCL sends Unconfirmed frame to ED                  |   |  | ← | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 1 (enable)<br>Back-off = 0<br>SmartEnableLevel = 0<br>SecondChIdx = 1<br>SecondChDr = FSK (invalid) [2]<br>SecondChAckOffset = Relay's second channel ACKOffset<br>SecondChFreq = Relay's second channel freq<br>Payload = [0x]41XXXXXXXXXXXX                                  | SecondChDr = FSK                         |
| 13 | If the Relay belongs to a region that supports FSK [2], ED must send Unconfirmed frame to TCL directly     |   |  | → | MAC-CMD EndDeviceConfAns<br>Payload NOT = [0x]410F                                                                                                                                                                                                                                                                                            |                                          |
|    | TCL sends Unconfirmed frame to ED                                                                          |   |  | ← | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 1 (enable)<br>Back-off = 0<br>SmartEnableLevel = 0<br>SecondChIdx = 2 (invalid)<br>SecondChDr = Relay's second channel DR<br>SecondChAckOffset = Invalid out of band value [2]<br>SecondChFreq = Relay's second channel Freq<br>Payload = [0x]41XXXXXXXXXXXX                   | Invalid SecondChIdx                      |
| 14 | ED must send Unconfirmed frame to TCL directly                                                             |   |  | → | MAC-CMD EndDeviceConfAns<br>Payload NOT = [0x]410F                                                                                                                                                                                                                                                                                            |                                          |
| 15 | ED must send Unconfirmed frame to TCL directly                                                             |   |  | → |                                                                                                                                                                                                                                                                                                                                               |                                          |
|    | TCL sends Unconfirmed frame to ED                                                                          |   |  | ← | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 1 (enable)<br>Back-off = 0<br>SmartEnableLevel = 0<br>SecondChIdx = 0 (disable second channel)<br>SecondChDr = Relay's second channel DR<br>SecondChAckOffset = Relay's second channel ACKOffset<br>SecondChFreq = Relay's second channel freq<br>Payload = [0x]41XXXXXXXXXXXX | Enable Relay but disable second channel  |
| 16 | ED must send Unconfirmed frame to the Relay using the default channel settings                             | → |  |   | PHYPayload includes<br>MAC-CMD EndDeviceConfAns<br>Payload = [0x]41XF                                                                                                                                                                                                                                                                         | End device used default channel settings |

|    |                                                                                               |   |   |   |                                                                                                                                                                                                                                                                                                                                                                           |                        |
|----|-----------------------------------------------------------------------------------------------|---|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
|    | Relay sends an Unconfirmed frame to the TCL                                                   |   | → |   | PHYPayload includes<br>MAC-CMD EndDeviceConfAns<br>Payload = [0x]41XF<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXX                                                                                                                                                                                |                        |
|    | TCL sends Unconfirmed frame to Relay                                                          |   | ← |   | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 1 (500ms)<br>DefaultChIdx = second default channel for the region if supported, else default channel [2]<br>SecondChIdx = 0<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 1 (200kHz)<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXXXXX | Disable second channel |
| 17 | Relay sends an Unconfirmed frame to the TCL                                                   |   | → |   | MAC-CMD RelayConfAns<br>Status = 1 for all ACK bits<br>Payload = [0x]40XX<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                                                  |                        |
| 18 | ED must send Unconfirmed frame to the Relay using the default channel (DefaultChIdx) settings | → |   |   | <i>PHYPayload</i><br><br>Note: Multiple uplinks may be received from the ED before receiving an uplink from the ED through the Relay                                                                                                                                                                                                                                      |                        |
|    | Relay sends an Unconfirmed frame to the TCL                                                   |   | → |   | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXX                                                                                                                                                                                                                                                             |                        |
| 19 | ED must send Unconfirmed frame to TCL directly                                                |   |   | → |                                                                                                                                                                                                                                                                                                                                                                           |                        |
|    | TCL sends Unconfirmed frame to ED                                                             |   |   | ← | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 0 (disable)<br>Back-off = 0<br>SmartEnableLevel = 0<br>SecondChIdx = 0 (disable second channel)<br>SecondChDr = Relay's second channel DR<br>SecondChAckOffset = Relay's second channel ACKOffset<br>SecondChFreq = Relay's second channel freq<br>Payload = [0x]41XXXXXXXXXXXXX                           |                        |
| 20 | ED must send Unconfirmed frame to the TCL                                                     |   |   | → | MAC-CMD EndDeviceConfAns<br>Payload = [0x]41XF                                                                                                                                                                                                                                                                                                                            |                        |

|    |                                                 |           |   |   |                                                                                                                                                                                                                                                                                                                        |  |
|----|-------------------------------------------------|-----------|---|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 21 | Relay must send an Unconfirmed frame to the TCL |           | → |   |                                                                                                                                                                                                                                                                                                                        |  |
|    | TCL sends Unconfirmed frame to Relay            |           | ← |   | MAC-CMD CtrlUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>CtrlUplinkAction = 0<br>Payload = [0x]44XX                                                                                                                                                                                                              |  |
| 22 | Relay must send an Unconfirmed frame to the TCL |           | → |   | MAC-CMD CtrlUplinkListAns<br>UplinkListIdxACK = 1<br>WFCnt32 = next valid value expected by the Relay<br>Payload = [0x]44XXXXXXXXXX                                                                                                                                                                                    |  |
| 23 | ED must send Unconfirmed frame                  |           |   | → |                                                                                                                                                                                                                                                                                                                        |  |
|    | TCL sends an Unconfirmed frame to ED            |           |   | ← | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 1 (enable)<br>Back-off = 5<br>SmartEnableLevel = 0<br>SecondChIdx = 0<br>SecondChDr = Relay's second channel DR<br>SecondChAckOffset = Relay's second channel ACKoffset<br>SecondChFreq = Relay's second channel freq<br>Payload = [0x]41XXXXXXXXXXXXXX |  |
| 24 | ED must send Unconfirmed frame to the Relay     | →         |   |   | PHYPayload includes<br>MAC-CMD EndDeviceConfAns<br>Payload = [0x]41XF                                                                                                                                                                                                                                                  |  |
|    | Relay must send Unconfirmed frame to TCL        |           | → |   | PHYPayload includes<br>MAC-CMD EndDeviceConfAns<br>Payload = [0x]41XF<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXX                                                                                                                            |  |
|    | TCL sends Unconfirmed frame to Relay            |           | ← |   | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 3<br>UplinkLimitReloadRate = 63<br>RootWorSKey of ED<br>DevAddr of ED<br>WFCnt32 = WFCnt32 + 10<br>Payload = [0x]43 XX (26 octets)                                                                                            |  |
| 25 | Relay must send Unconfirmed frame to the TCL    |           | → |   | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                         |  |
| 26 | ED must send WOR frames to Relay                | →<br>R[5] |   |   |                                                                                                                                                                                                                                                                                                                        |  |

|    |                                                                                |   |   |      |                                                                                                                                                                                                                                                                                                                      |                                                                           |
|----|--------------------------------------------------------------------------------|---|---|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|    | Relay sends an Unconfirmed frame to the TCL (tolerance of 5%)                  |   | → | R[5] | MAC-CMD NotifyNewEndDeviceReq<br>DevAddr of ED<br>Payload = [0x]46XXXXXXXXXXXX                                                                                                                                                                                                                                       | Any NotifyEndDeviceReq may be lost because the Relay is sending an uplink |
| 27 | ED must send Unconfirmed frame to TCL                                          |   |   | →    |                                                                                                                                                                                                                                                                                                                      | Sent via the Gateway only                                                 |
| 28 | ED must send Unconfirmed frame to the Relay                                    | → |   |      | <i>PHYPayload</i>                                                                                                                                                                                                                                                                                                    | Sent via the Gateway and Relay                                            |
|    | Relay sends an Unconfirmed frame to the TCL                                    |   | → |      | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                         | Note: some NotifyNewEndDeviceReq may be sent previously                   |
| 29 | Relay must send Unconfirmed frame to TCL                                       |   | → |      |                                                                                                                                                                                                                                                                                                                      |                                                                           |
|    | TCL sends Unconfirmed frame to Relay                                           |   | ← |      | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 3<br>UplinkLimitReloadRate = 63<br>RootWorSKey of ED<br>DevAddr of ED<br>WFCnt32 = 0<br>Payload = [0x]43 XX (26 octets)                                                                                                     | Revert to default                                                         |
| 30 | Relay must send Unconfirmed frame to the TCL                                   |   | → |      | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                       |                                                                           |
| 31 | ED must send Unconfirmed frame to the Relay using the default channel settings | → |   |      | <i>PHYPayload</i>                                                                                                                                                                                                                                                                                                    |                                                                           |
|    | Relay sends an Unconfirmed frame to the TCL                                    |   | → |      | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                         |                                                                           |
|    | TCL sends an Unconfirmed frame to ED                                           |   | ← |      | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 1 (enable)<br>Back-off = 0<br>SmartEnableLevel = 0<br>SecondChIdx = 0<br>SecondChDr = Relay's second channel DR<br>SecondChAckOffset = Relay's second channel ACKoffset<br>SecondChFreq = Relay's second channel freq<br>Payload = [0x]41XXXXXXXXXXXX |                                                                           |

|    |                                                                                               |   |   |  |                                                                                                                                                                                                                                                                                                                                 |                             |
|----|-----------------------------------------------------------------------------------------------|---|---|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| 32 | ED must send Unconfirmed frame to the Relay using the default channel settings                | → |   |  | PHYPayload includes<br>MAC-CMD EndDeviceConfAns<br>Payload = [0x]41XF                                                                                                                                                                                                                                                           |                             |
|    | Relay sends an Unconfirmed frame to the TCL                                                   |   | → |  | PHYPayload includes<br>MAC-CMD EndDeviceConfAns<br>Payload = [0x]41XF<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                       |                             |
| 33 | Relay sends an Unconfirmed frame to the TCL                                                   |   | → |  |                                                                                                                                                                                                                                                                                                                                 |                             |
|    | TCL sends Unconfirmed frame to Relay                                                          |   | ← |  | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 0 (1s)<br>DefaultChIdx = first default channel for the region [2]<br>SecondChIdx = 0<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 1 (200kHz)<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX |                             |
| 34 | Relay sends an Unconfirmed frame to the TCL                                                   |   | → |  | MAC-CMD RelayConfAns<br>Status = 1 for all ACK bits<br>Payload = [0x]40XX<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                        |                             |
| 35 | ED must send Unconfirmed frame to the Relay using the default channel (DefaultChIdx) settings | → |   |  | PHYPayload<br><br>Note: Multiple uplinks may be received from the ED before receiving an uplink from the ED through the Relay                                                                                                                                                                                                   |                             |
|    | The TCL sends Unconfirmed frames                                                              |   | ← |  | PHYPayload includes<br>Confirmed downlink<br>CP-CMD RegionalDutyCycleCtrlReq-ON<br>FPort = 224<br>Payload = [0x]0501<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                        | Turn ON Regional Duty Cycle |
| 36 | ED sends an Unconfirmed frame to the Relay                                                    | → |   |  | PHYPayload includes<br>ACK bit = 1                                                                                                                                                                                                                                                                                              |                             |

|  |                                             |  |   |  |                                                                                                                                                        |  |
|--|---------------------------------------------|--|---|--|--------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | Relay sends an Unconfirmed frame to the TCL |  | → |  | PHYPayload includes<br>ACK bit = 1<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX |  |
|--|---------------------------------------------|--|---|--|--------------------------------------------------------------------------------------------------------------------------------------------------------|--|

313

## 314 2.3. Other End Device tests

### 315 **Max Payload test**

- 316 • TCL sends an EchoPayloadReq to the ED via Relay for MaxPayload for Repeater for Max125kHzDR [2]
- 317
- 318 • ED/Relay must respond with EchoPayloadAns
- 319 • TCL sends an EchoPayloadReq to the ED via Relay exceeding MaxPayload (MaxPayloadLength for Repeater + 2) for Max125kHzDR
- 320
- 321 • ED/Relay PHYPayload must not exceed limits
- 322
- 323

### 324 **Confirmed uplink - downlink tests**

- 325 • When ED sends an Unconfirmed frame to the TCL via Relay, TCL sends an Unconfirmed frame to the Relay where PHYPayload includes a Confirmed downlink to ED; and with CP-CMD TxFramesCtrlReq to request the ED to switch to Confirmed frames.
- 326
- 327
- 328 • Relay must forward the ED response to the TCL, and ED sends a Confirmed frame with a ACK Bit = True for the Confirmed downlink.
- 329
- 330 • TCL then sends an Unconfirmed frame with CP-CMD TxFramesCtrlReq to revert to Unconfirmed frames.
- 331
- 332
- 333

## 334 Test Procedure Frame Sequence Chart

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336

| Step | Procedure                                   | ED<br>↔<br>R | R<br>↔<br>TCL | ED<br>↔<br>TCL | Frame                                                                                                                                                                                                                                 | Remarks                                             |
|------|---------------------------------------------|--------------|---------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 1    | ED sends Unconfirmed frame to Relay         | →            |               |                | <i>PHYPayload</i>                                                                                                                                                                                                                     |                                                     |
|      | Relay sends an Unconfirmed frame to the TCL |              | →             |                | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                          |                                                     |
|      | TCL sends Unconfirmed frame to Relay        |              | ←             |                | PHYPayload includes CP-CMD EchoPayloadReq<br>FPort = 224<br>Payload = [0x]08XX...<br>(MaxPayloadLength for Repeater for Max125kHzDR, refer [2])<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226                                 | Relay sends echo request to ED for MaxPayload       |
| 2    | ED sends Unconfirmed frame to Relay         | →            |               |                | <i>PHYPayload includes CP-CMD EchoPayloadAns</i><br>FPort = 224<br>Payload = [0x]08XX...<br>(MaxPayloadLength for Repeater for Max125kHzDR, refer [2])                                                                                |                                                     |
|      | Relay sends an Unconfirmed frame to the TCL |              | →             |                | PHYPayload includes CP-CMD EchoPayloadAns<br>FPort = 224<br>Payload = [0x]08XX...<br>(MaxPayloadLength for Repeater for Max125kHzDR, refer [2])<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226 | Max Payload                                         |
| 3    | ED sends Unconfirmed frame to Relay         | →            |               |                | <i>PHYPayload</i>                                                                                                                                                                                                                     |                                                     |
|      | Relay sends an Unconfirmed frame to the TCL |              | →             |                | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                          |                                                     |
|      | TCL sends Unconfirmed frame to Relay        |              | ←             |                | PHYPayload includes CP-CMD EchoPayloadReq<br>FPort = 224<br>Payload = [0x]08XX...<br>(MaxPayloadLength for Repeater + 2 for Max125kHzDR, refer [2])<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226                             | Relay sends echo request to ED exceeding MaxPayload |
| 4    | ED sends Unconfirmed frame to Relay         | →            |               |                | <i>PHYPayload does not exceed limits</i>                                                                                                                                                                                              | Payload does not exceed limits                      |
|      | Relay sends an Unconfirmed frame to the TCL |              | →             |                | PHYPayload does not exceed limits<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226                                                                                                               |                                                     |
| 5    | ED sends Unconfirmed frame to Relay         | →            |               |                | <i>PHYPayload</i>                                                                                                                                                                                                                     |                                                     |

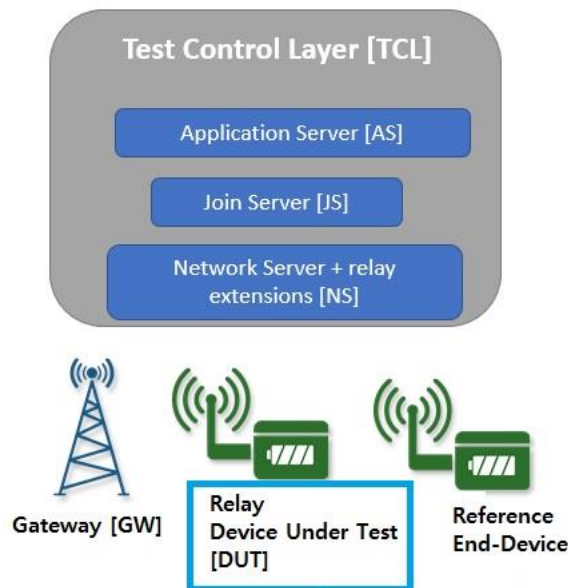
|   |                                             |   |   |  |                                                                                                                                                                                                  |  |
|---|---------------------------------------------|---|---|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|   | Relay sends an Unconfirmed frame to the TCL |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226                                                                                                                   |  |
|   | TCL sends Unconfirmed frame to Relay        |   | ← |  | PHYPayload includes a Confirmed downlink to ED<br>CP-CMD TxFramesCtrlReq<br>FPort = 224<br>Frame type = Confirmed<br>Payload = [0x]0702<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226    |  |
| 6 | ED sends Confirmed frame to Relay           | → |   |  | <i>PHYPayload includes ACK Bit = True</i><br><br>Note: The ED may split this frame into first an empty Unconfirmed frame with ACK, followed by a Confirmed frame. This must be accepted as well. |  |
|   | Relay sends an Unconfirmed frame to the TCL |   | → |  | PHYPayload includes<br><i>ACK Bit = True</i><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Uplink = Confirmed frame<br>Payload = [0x]XXXXXXXXXXXX         |  |
|   | TCL sends Unconfirmed frame to Relay        |   | ← |  | PHYPayload includes<br>Acknowledge<br>CP-CMD TxFramesCtrlReq<br>FPort = 224<br>Frame type = Unconfirmed<br>Payload = [0x]0701<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226              |  |
| 7 | ED sends Unconfirmed frame to Relay         | → |   |  | <i>PHYPayload</i>                                                                                                                                                                                |  |
|   | Relay sends an Unconfirmed frame to the TCL |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                     |  |



### 3. DUT = RELAY

The tests are conducted in a test harness generally comprised of the components represented in Figure 2.

**Note:** The forward limitation for the Relay must be disabled when running all tests in Section 3, except ConfigureFwdLimitReq test.



**Figure 2: Test Harness Architecture when the DUT = Relay device**

#### 3.1. Pre-Requisite Setup Test

The Pre-requisite setup tests need to be run as a first step on the Relay and ED before performing any of the other test cases mentioned in this section, to setup the devices and speed up the test execution.

The Pre-requisite tests include-

##### Activation of DUT Relay device (Relay)

- Standard activation (OTAA/ABP)
- Relay sends an uplink after activation
- Change TXPeriodicity to 5 seconds
- Switch to Unconfirmed uplink frames (if not already)
- Turn on the ADR bit (if not already set)
- Change the Relay uplink DataRate to Max125KHzDR
- Enable the Relay with RelayConfReq command
- Set the relay to no limit with ConfigureFwdLimitReq command

361 - Change the Relay uplink periodicity to 30s

362

363 **Activation of ED1 (Reference End-Device)**

364 - ED sends Join-Request to TCL

365 - TCL responds with Join-Accept

366 - Change TXPeriodicity to 5 seconds

367 - TCL enables the Relay mode on the ED

368 - ED sends the Join-Request to the Relay

369 - Relay forwards the Join-Request to the TCL

370 - TCL responds with Join-Accept

371 - Relay sends empty uplink to the TCL

372 - TCL does not respond with UpdateUplinkListReq to the Relay

373 - Relay sends NotifyNewEndDeviceReq to the TCL

374 - TCL sends UpdateUplinkListReq to the Relay

375 - Change TXPeriodicity to 20 seconds

376 - Switch to Unconfirmed uplink frames (if not already)

377 - Turn on the ADR bit (if not already set)

378 - Change the uplink DataRate to Max125KHz

379 - TCL sends the EndDeviceConfReq to the ED via the Relay to enable the Relay mode on the ED

381 - ED responds with the EndDeviceConfAns via the Relay

382 -

383

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385 **Test Procedure Frame Sequence Chart**

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| Step | Procedure                                                                |               |                |                 | Frame Sequence                                                                                                                                    | Remarks                             |
|------|--------------------------------------------------------------------------|---------------|----------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
|      | <b>ACTIVATION of DUT RELAY DEVICE</b>                                    | ED<br><><br>R | R<br><><br>TCL | ED<br><><br>TCL |                                                                                                                                                   |                                     |
| 1    | If Relay = OTAA device, Relay must send a Join-Request frame             |               | →              |                 |                                                                                                                                                   |                                     |
|      | If Relay = ABP device, skip Step 1.<br>TCL sends a Join-Accept frame     |               | ←              |                 |                                                                                                                                                   |                                     |
| 2    | Relay must send Confirmed or Unconfirmed frame                           |               | →              |                 |                                                                                                                                                   |                                     |
|      | The TCL sends Unconfirmed frame to Relay                                 |               | ←              |                 | CP-CMD<br>TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 5<br>Payload = [0x]0601<br><br>If Confirmed uplink, Acknowledge                  | Change the Periodicity of the Relay |
| 3    | Relay must send Confirmed or Unconfirmed frame                           |               | →              |                 |                                                                                                                                                   |                                     |
|      | If Relay sent Confirmed frame, then TCL sends Unconfirmed frame to Relay |               | ←              |                 | Acknowledge<br><br>CP-CMD TxFramesCtrlReq<br>Frame type = Unconfirmed<br>Payload = [0x]0701                                                       |                                     |
|      | Else, skip this step                                                     |               |                |                 |                                                                                                                                                   |                                     |
| 4    | Relay sends Unconfirmed frame                                            |               | →              |                 |                                                                                                                                                   |                                     |
|      | TCL sends Unconfirmed frame to Relay                                     |               | ←              |                 | If FCtrl ADR Bit = false, then perform this step, else skip to next downlink<br><br>CP-CMD AdrBitChangeReq-ON<br>FPort = 224<br>Payload = [0x]040 |                                     |
| 5    | Relay must send Unconfirmed frame                                        |               | →              |                 |                                                                                                                                                   |                                     |

|                                            |                                                 |  |   |   |                                                                                                                                                                                                                                                                                                                               |                                     |
|--------------------------------------------|-------------------------------------------------|--|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
|                                            | The TCL sends Unconfirmed frame to Relay        |  | ← |   | MAC-CMD LinkADRReq<br>FPort = 0<br>DataRate = Max125kHzDR, refer [2]<br>ChMaskCntl:<br>DC = 0,<br>FC = 6<br>ChMask:<br>DC - Enable only default channels<br>FC = [0x]00FF<br>Payload = [0x]03XXXXXXXXXX                                                                                                                       |                                     |
| 6                                          | Relay must send Unconfirmed frame to TCL        |  | → |   | MAC-CMD LinkADRAns<br>Payload = [0x]0307                                                                                                                                                                                                                                                                                      |                                     |
|                                            | TCL sends Unconfirmed frame to Relay            |  | ← |   | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 0 (1s)<br>DefaultChIdx = any channel allowed for the region [2]<br>SecondChIdx = 0<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 1 (200kHz)<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX |                                     |
| 7                                          | Relay must send an Unconfirmed frame to the TCL |  | → |   | MAC-CMD RelayConfAns<br>Status = 1 for all ACK bits<br>Payload = [0x]40XX                                                                                                                                                                                                                                                     |                                     |
|                                            | TCL sends Unconfirmed frame to Relay            |  | ← |   | MAC-CMD ConfigureFwdLimitReq<br>FPort = 0<br>ResetLimitCounter = 1 (reload rate)<br>JoinReqReloadRate = 127<br>NotifyReloadRate = 127<br>GlobalUplinkReloadRate = 127<br>OverallReloadRate = 127<br>FwdLimitLoadCapacityPL = max size<br>Payload = [0x]45XXXXXXXXXX                                                           |                                     |
| 8                                          | Relay must send an Unconfirmed frame to the TCL |  | → |   | MAC-CMD ConfigureFwdLimitAns<br>Payload = [0x]45                                                                                                                                                                                                                                                                              |                                     |
|                                            | TCL sends Unconfirmed frame to Relay            |  | ← |   | CP-CMD TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 30<br>Payload = [0x]0604                                                                                                                                                                                                                                        | Change the Periodicity of the Relay |
| <b>ACTIVATION of REFERENCE END-DEVICE1</b> |                                                 |  |   |   |                                                                                                                                                                                                                                                                                                                               |                                     |
| 9                                          | ED sends a Join-Request frame                   |  |   | → |                                                                                                                                                                                                                                                                                                                               |                                     |

|    |                                                    |   |   |   |                                                                                                                                                                                                                     |  |
|----|----------------------------------------------------|---|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    | TCL sends a Join-Accept frame                      |   |   | ← |                                                                                                                                                                                                                     |  |
| 10 | ED sends Confirmed or Unconfirmed frame            |   |   | → |                                                                                                                                                                                                                     |  |
|    | TCL sends Unconfirmed frame                        |   |   | ← | If ED sent Confirmed uplink frame, TCL must Acknowledge<br><br>CP-CMD<br>TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 5<br>Payload = [0x]0601                                                             |  |
| 11 | ED sends Confirmed or Unconfirmed frame            |   |   | → |                                                                                                                                                                                                                     |  |
|    | TCL sends Unconfirmed frame                        |   |   | ← | If ED sent Confirmed uplink frame, TCL must Acknowledge<br><br>CP-CMD RelayModeCtrl<br>Ctrl = 1<br>FPort = 224<br>Payload = [0x]53XX                                                                                |  |
| 12 | ED sends Confirmed or Unconfirmed frame            |   |   | → |                                                                                                                                                                                                                     |  |
|    | TCL sends Unconfirmed frame                        |   |   | ← | If ED sent Confirmed uplink frame, TCL must Acknowledge<br><br>CP-CMD DutJoinReq<br>FPort = 224<br>Payload = [0x]02                                                                                                 |  |
| 13 | ED sends Unconfirmed frame to Relay                | → |   |   | <i>WORRelayJoin and PHYPayload which includes the Join-Request</i>                                                                                                                                                  |  |
|    | Relay must send Unconfirmed frame to the TCL       |   | → |   | ForwardUplinkReq (UplinkMetaData, UplinkFrequency, PHYPayload containing Join-Request)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXX                                                                           |  |
|    | TCL sends Unconfirmed frame to Relay               |   | ← |   | ForwardDownlinkReq (PHYPayload containing Join-Accept)<br>FPort = 226<br>Payload = [0x]variable                                                                                                                     |  |
| 14 | Relay must send Unconfirmed frame to the TCL       |   | → |   | Empty uplink                                                                                                                                                                                                        |  |
|    | TCL sends Unconfirmed frame to Relay on RX2 window |   | ← |   | MAC-CMD<br>UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 3<br>UplinkLimitReloadRate = 63<br>RootWorSKey of ED<br>DevAddr of ED<br>WFCnt32 = 0<br>Payload = [0x]43 XX (26 octets) |  |

|    |                                                        |   |   |   |                                                                                                                                                                                                                                                                                       |                                        |
|----|--------------------------------------------------------|---|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 15 | Relay must send Unconfirmed frame to the TCL           |   | → |   | MAC-CMD<br>UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                     |                                        |
| 16 | ED sends Confirmed or Unconfirmed frame                |   |   | → |                                                                                                                                                                                                                                                                                       |                                        |
|    | TCL sends Unconfirmed frame to ED                      |   |   | ← | CP-CMD<br>TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 20<br>Payload = [0x]0603<br><br>If ED uplink was a Confirmed frame, send ACK                                                                                                                                         |                                        |
| 17 | ED sends a Confirmed or Unconfirmed frame to the Relay | → |   |   | PHYPayload                                                                                                                                                                                                                                                                            |                                        |
|    | Relay must send Unconfirmed frame to TCL               |   | → |   | ForwardUplinkReq<br>(UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                       |                                        |
|    | The TCL sends Unconfirmed frame to the Relay           |   | ← |   | PHYPayload includes<br>CP-CMD TxFramesCtrlReq<br>FPort = 224<br>Frame type = Unconfirmed<br>Payload = [0x]0701<br>If ED uplink was a Confirmed frame, TCL must acknowledge and perform this step, else skip to next downlink<br><br>ForwardDownlinkReq<br>(PHYPayload)<br>FPort = 226 | Relay then sends downlink to ED on RXR |
| 18 | ED must send an Unconfirmed frame to the Relay         | → |   |   | PHYPayload                                                                                                                                                                                                                                                                            |                                        |
|    | Relay sends Unconfirmed frame to TCL                   |   | → |   | ForwardUplinkReq<br>(UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                       |                                        |

|    |                                              |   |   |  |                                                                                                                                                                                                                                                                                                                                                                         |                                        |
|----|----------------------------------------------|---|---|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | <p>If FCtrl ADR Bit = false, then perform this step, else skip to next downlink</p> <p>PHYPayload includes<br/>CP-CMD AdrBitChangeReq-ON<br/>FPort = 224<br/>Payload = [0x]040</p> <p>ForwardDownlinkReq (PHYPayload)<br/>FPort = 226</p>                                                                                                                               | Relay then sends downlink to ED on RXR |
| 19 | ED sends an Unconfirmed frame to the Relay   | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                                                                                              |                                        |
|    | Relay must send Unconfirmed frame to TCL     |   | → |  | <p>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br/>FPort = 226<br/>Payload = [0x]XXXXXXXXXXXX</p>                                                                                                                                                                                                                                                   |                                        |
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | <p>PHYPayload includes<br/>MAC-CMD LinkADRReq<br/>TXPower = Maximum, refer <a href="#">[2]</a><br/>DataRate = Max125kHzDR, refer <a href="#">[2]</a><br/>Payload = [0x]03XXXXXXXX<br/>ChMaskCntl:<br/>    DC = 0,<br/>    FC = 6<br/>ChMask:<br/>    DC - Enable only default channels<br/>    FC = [0x]00FF</p> <p>ForwardDownlinkReq (PHYPayload)<br/>FPort = 226</p> | Relay then sends downlink to ED on RXR |
| 20 | ED sends an Unconfirmed frame to the Relay   | → |   |  | <p>PHYPayload includes<br/>MAC-CMD LinkADRAns<br/>Payload = [0x]0307</p>                                                                                                                                                                                                                                                                                                |                                        |
|    | Relay must send Unconfirmed frame to TCL     |   | → |  | <p>PHYPayload includes<br/>MAC-CMD LinkADRAns<br/>Payload = [0x]0307</p> <p>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br/>FPort = 226<br/>Payload = [0x]XXXXXXXXXXXX</p>                                                                                                                                                                          |                                        |

|    |                                              |   |   |  |                                                                                                                                                                                                                                                                                                                                                                              |              |
|----|----------------------------------------------|---|---|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | PHYPayload includes<br>MAC-CMD<br>EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 1 (enable)<br>Back-off = 0<br>SecondChIdx = 0<br>SecondChDr = Relay's second channel DR<br>SecondChAckOffset = Relay's second channel ACKOffset<br>SecondChFreq = Relay's second channel freq<br>Payload = [0x]41XXXXXXXXXXXX<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226 |              |
| 21 | ED sends an Unconfirmed frame to the Relay   | → |   |  | <i>PHYPayload includes</i><br>MAC-CMD<br>EndDeviceConfAns<br>Payload = [0x]41XF                                                                                                                                                                                                                                                                                              | Enable Relay |
|    | Relay sends Unconfirmed frame to TCL         |   | → |  | PHYPayload includes<br>MAC-CMD<br>EndDeviceConfAns<br>Payload = [0x]41XF<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                 |              |

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### 3.2. NotifyNewEndDeviceReq Test

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- TCL sends the UpdateUplinkListReq with an INVALID WFCnt32 but the same DevAddr and RootWorSKey to the Relay.
- Relay responds with UpdateUplinkListAns
- ED sends and uplink to the Relay and the Relay forwards it to the TCL
- TCL sends CtrlUplinkListReq to remove Index = 1 from the trusted list
- Relay responds with CtrlUplinkListAns
- ED sends and uplink to the Relay and the Relay does not forward it to the TCL but instead sends a NotifyNewEndDeviceReq to the TCL
- TCL sends the UpdateUplinkListReq with a valid WFCnt32 = next expected value to the Relay.
- Relay responds with UpdateUplinkListAns





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- 401       • ED sends and uplink to the Relay and the Relay forwards it to the TCL  
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403



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## 404 Test Procedure Frame Sequence Chart

| Step | Procedure                                       |               |            |                 | Frame Sequence                                                                                                                                                                                                           | Test Purpose                                |
|------|-------------------------------------------------|---------------|------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
|      |                                                 | ED<br><><br>R | R<><br>TCL | ED<br><><br>TCL | Frame                                                                                                                                                                                                                    |                                             |
| 1    | Relay sends Unconfirmed frame to the TCL        |               | →          |                 |                                                                                                                                                                                                                          |                                             |
|      | TCL sends Unconfirmed frame to Relay            |               | ←          |                 | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 2<br>UplinkLimitBucketSize = 3<br>UplinkLimitReloadRate = 63<br>RootWorSKey of ED<br>WFCnt32 = INCORRECT VALUE<br>DevAddr<br>Payload = [0x]43 XX (26 octets) | ED with invalid WFCnt added to trusted list |
| 2    | Relay sends Unconfirmed frame to the TCL        |               | →          |                 | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                           |                                             |
| 3    | ED sends an Unconfirmed frame to the Relay      | →             |            |                 | PHYPayload                                                                                                                                                                                                               |                                             |
|      | Relay must send Unconfirmed frame to TCL        |               | →          |                 | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                             | Relay forwards uplink for Index = 1         |
|      | The TCL sends Unconfirmed frame to the Relay    |               | ←          |                 | PHYPayload includes<br>MAC-CMD CtrlUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>CtrlUplinkAction = 1<br>Payload = [0x]44XX<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226                                   | Remove Index = 1                            |
| 4    | Relay must send an Unconfirmed frame to the TCL |               | →          |                 | MAC-CMD CtrlUplinkListAns<br>UplinkListIdxACK = 1<br>WFCnt32 = next valid value expected by the Relay<br>Payload = [0x]44XXXXXXXXXX                                                                                      |                                             |
| 5    | ED sends an Unconfirmed frame to the Relay      | →             |            |                 | PHYPayload                                                                                                                                                                                                               |                                             |
|      | Relay must send Unconfirmed frame to TCL        |               | →          |                 | MAC-CMD NotifyNewEndDeviceReq<br>DevAddr of ED<br>Payload = [0x]46XXXXXXXXXXXX                                                                                                                                           | Relay sends NotifyNewEndDeviceReq           |

|   |                                            |   |   |  |                                                                                                                                                                                                                              |                                                       |
|---|--------------------------------------------|---|---|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
|   | TCL sends Unconfirmed frame to Relay       |   | ← |  | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 3<br>UplinkLimitBucketSize = 3<br>UplinkLimitReloadRate = 63<br>RootWorSKey of ED<br>WFCnt32 = next expected value<br>DevAddr<br>Payload = [0x]43 XX (26 octets) | TCL adds Index = 3 to trusted list with correct value |
| 6 | Relay sends Unconfirmed frame to the TCL   |   | → |  | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                               |                                                       |
| 7 | ED sends an Unconfirmed frame to the Relay | → |   |  | <i>PHYPayload</i>                                                                                                                                                                                                            |                                                       |
|   | Relay must send Unconfirmed frame to TCL   |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                 | Relay forwards uplink for Index = 3                   |

### 3.3. Relay MAC commands

#### 3.3.1 RelayConfReq

- Disable the Duty cycle using the CP-CMD RegionalDutyCycleCtrlReq
- TCL sends the Relay a RelayConfReq MAC command with StartStop = 0 (disable)
- Relay must be disabled
- When ED sends the next uplink to the Relay, the Relay must not forward it to the TCL
- TCL sends the Relay a RelayConfReq MAC command with StartStop = 1 (enable)
  - StartStop = 1
  - CADPeriodicity = 1 (500ms),
  - DefaultChIdx = a different channel allowed for the region [2],
  - SecondChIdx = 0,
  - SecondChDr = a DR allowed for the region [2],
  - SecondChAckOffset = 1 (200kHz)
  - SecondChFreq = a frequency allowed for the region [2]
- Relay must be enabled and must respond with the RelayConfAns MAC command with Status = 1 for all the ACK bits.
- ED sends an uplink to the Relay using the default newly configured.
- The Relay must forward the uplink to the TCL using the new parameters set.

#### Invalid parameters tests

- TCL sends the Relay a RelayConfReq MAC command with
  - CADPeriodicity = 1 (500ms),
  - DefaultChIdx = a channel NOT allowed for the region [2], (invalid)
  - SecondChIdx = 0,
  - SecondChDr = a DR allowed for the region [2],
  - SecondChAckOffset = 0,
  - SecondChFreq = a frequency allowed for the region [2]
- Relay must respond with RelayConfAns with Payload NOT = [0x]403F
- TCL sends the Relay a RelayConfReq MAC command with
  - CADPeriodicity = 0 (1s),
  - DefaultChIdx = a channel allowed for the region [2],
  - SecondChIdx = 2, (invalid)
  - SecondChDr = a DR allowed for the region [2],
  - SecondChAckOffset = 0,
  - SecondChFreq = a frequency allowed for the region [2]
- Relay must respond with RelayConfAns with Payload NOT = [0x]403F
- TCL sends the Relay a RelayConfReq MAC command with
  - CADPeriodicity = 1 (500ms),
  - DefaultChIdx = a channel allowed for the region [2],
  - SecondChIdx = 1,
  - SecondChDr = a DR NOT allowed for the region [2], (invalid)
  - SecondChAckOffset = 0,
  - SecondChFreq = a frequency allowed for the region [2]
- Relay must respond with RelayConfAns with Payload NOT = [0x]403F
- If the Relay belongs to a region that supports LR-FHSS [2], TCL sends the Relay a RelayConfReq MAC command with
  - CADPeriodicity = 1 (500ms),
  - DefaultChIdx = a channel allowed for the region [2],
  - SecondChIdx = 1,
  - SecondChDr = LR-FHSS DR [2], (invalid)
  - SecondChAckOffset = 0,
  - SecondChFreq = a frequency allowed for the region [2]
- Relay must respond with RelayConfAns with Payload NOT = [0x]403F
- If the Relay belongs to a region that supports FSK [2], TCL sends the Relay a RelayConfReq MAC command with
  - CADPeriodicity = 1 (500ms),
  - DefaultChIdx = a channel allowed for the region [2],

- 469                   ○ SecondChIdx = 1,
- 470                   ○ SecondChDr = FSK [2], (invalid)
- 471                   ○ SecondChAckOffset = 0,
- 472                   ○ SecondChFreq = a frequency allowed for the region [2]
- 473       • Relay must respond with RelayConfAns with Payload NOT = [0x]403F
- 474
- 475       • TCL sends the Relay a RelayConfReq MAC command with
- 476                   ○ CADPeriodicity = 1 (500ms),
- 477                   ○ DefaultChIdx = a channel allowed for the region [2],
- 478                   ○ SecondChIdx = 1,
- 479                   ○ SecondChDr = a DR allowed for the region [2],
- 480                   ○ SecondChAckOffset = 6, (invalid)
- 481                   ○ SecondChFreq = a frequency allowed for the region [2]
- 482       • Relay must respond with RelayConfAns with Payload NOT = [0x]403F
- 483
- 484       • TCL sends the Relay a RelayConfReq MAC command with
- 485                   ○ CADPeriodicity = 1 (500ms),
- 486                   ○ DefaultChIdx = a channel allowed for the region [2],
- 487                   ○ SecondChIdx = 1,
- 488                   ○ SecondChDr = a DR allowed for the region [2],
- 489                   ○ SecondChAckOffset = 0,
- 490                   ○ SecondChFreq = a frequency NOT allowed for the region [2] (invalid)
- 491       • Relay must respond with RelayConfAns with Payload NOT = [0x]403F
- 492
- 493       • TCL sends the Relay a RelayConfReq MAC command with
- 494                   ○ CADPeriodicity = out of band value (invalid),
- 495                   ○ DefaultChIdx = a channel allowed for the region [2],
- 496                   ○ SecondChIdx = 1,
- 497                   ○ SecondChDr = a channel allowed for the region [2],
- 498                   ○ SecondChAckOffset = 0,
- 499                   ○ SecondChFreq = a frequency allowed for the region [2]
- 500       • Relay must respond with RelayConfAns with Payload NOT = [0x]403F
- 501
- 502       • TCL sends the Relay a RelayConfReq MAC command with
- 503                   ○ CADPeriodicity = 0 (1s),
- 504                   ○ DefaultChIdx = a channel allowed for the region [2],
- 505                   ○ SecondChIdx = 0 (disable),
- 506                   ○ SecondChDr = a DR allowed for the region [2],
- 507                   ○ SecondChAckOffset = 1 (200kHz)
- 508                   ○ SecondChFreq = a frequency allowed for the region [2]
- 509       • Relay must respond with RelayConfAns all ACK bits set to 1



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- 510
- 511
- 512
- 513
- TCL sends the EndDeviceConfReq command to the ED to update the new Relay parameters
  - End-device sends the EndDeviceConfAns to the Relay using the default channel settings.
  - Relay forwards the uplink to the TCL.



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## 514 Test Procedure Frame Sequence Chart



| Step | Procedure                                    |              |           |                | Frame Sequence                                                                                                                                                                                                                                                                                                                           | Test Purpose                 |
|------|----------------------------------------------|--------------|-----------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
|      |                                              | ED<br>↔<br>R | R↔<br>TCL | ED<br>↔<br>TCL | Frame                                                                                                                                                                                                                                                                                                                                    |                              |
| 1    | ED sends an Unconfirmed frame to the Relay   | →            |           |                | PHYPayload                                                                                                                                                                                                                                                                                                                               |                              |
|      | Relay sends an Unconfirmed frame to the TCL  |              | →         |                | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                                             |                              |
|      | The TCL sends Unconfirmed frames             |              | ←         |                | PHYPayload includes<br>Confirmed downlink<br>CP-CMD RegionalDutyCycleCtrlReq-OFF<br>FPort = 224<br>Payload = [0x]0500<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                | Turn OFF Regional Duty Cycle |
| 2    | ED sends an Unconfirmed frame to the Relay   | →            |           |                | PHYPayload includes<br>ACK bit = 1                                                                                                                                                                                                                                                                                                       |                              |
|      | Relay must send Unconfirmed frame to TCL     |              | →         |                | PHYPayload includes<br>ACK bit = 1<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                   |                              |
|      | TCL sends Unconfirmed frame to Relay         |              | ←         |                | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 0<br>CADPeriodicity = 1 (500ms)<br>DefaultChIdx = a different channel allowed for the region [2]<br>SecondChIdx = 0<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 1 (200kHz)<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX | Disable Relay                |
| 3    | Relay sends an Unconfirmed frame to the TCL  |              | ←         |                | MAC-CMD RelayConfAns<br>Status = 1 for all ACK bits<br>Payload = [0x]40XX                                                                                                                                                                                                                                                                |                              |
| 4    | ED sends an Unconfirmed frame to the Relay   | →            |           |                | PHYPayload                                                                                                                                                                                                                                                                                                                               |                              |
|      | Relay must not forward this frame to the TCL |              |           |                |                                                                                                                                                                                                                                                                                                                                          | Relay is disabled            |
| 5    | Relay sends an Unconfirmed frame to the TCL  |              | ←         |                |                                                                                                                                                                                                                                                                                                                                          |                              |

|   |                                                                      |   |   |  |                                                                                                                                                                                                                                                                                                                                                                                                                                |                               |
|---|----------------------------------------------------------------------|---|---|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
|   | TCL sends Unconfirmed frame to Relay                                 |   | ← |  | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 1 (500ms)<br>DefaultChIdx = a different channel allowed for the region [2] (if supported)<br>SecondChIdx = 0<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 1 (200kHz)<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX                                                                        | Enable Relay                  |
| 6 | Relay sends an Unconfirmed frame to the TCL                          |   | ← |  | MAC-CMD RelayConfAns<br>Status = 1 for all ACK bits<br>Payload = [0x]40XX<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                                                                                                       |                               |
| 7 | ED sends an Unconfirmed frame to the Relay using the default channel | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                                                                                                                                                     |                               |
|   | Relay must send an Unconfirmed frame to the TCL                      |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                                                                                                                                   |                               |
|   | TCL sends Unconfirmed frame to Relay                                 |   | ← |  | Note: This command must be sent only for regions which support only 1 DefaultChIdx. Refer [2]<br><br>MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 1 (500ms)<br>DefaultChIdx = a channel NOT allowed for the region [2]<br>SecondChIdx = 0<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 0<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX | Invalid default channel index |
| 8 | Relay must send an Unconfirmed frame to the TCL                      |   | → |  | Note: This command must be sent only for regions which support only 1 DefaultChIdx. Refer [2]<br><br>MAC-CMD RelayConfAns<br>Payload NOT = [0x]403F<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                             |                               |
| 9 | ED sends an Unconfirmed frame to the Relay using the default channel | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                                                                                                                                                     |                               |
|   | Relay must send Unconfirmed frame to TCL                             |   | ← |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                                                                                                                                   |                               |

|    |                                                                                                  |   |   |  |                                                                                                                                                                                                                                                                                                                           |                              |
|----|--------------------------------------------------------------------------------------------------|---|---|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
|    | TCL sends Unconfirmed frame to Relay                                                             |   | ← |  | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 0 (1s)<br>DefaultChIdx = a channel allowed for the region [2]<br>SecondChIdx = 2<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 0<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX        | Invalid second channel index |
| 10 | Relay must send an Unconfirmed frame to the TCL                                                  |   | → |  | MAC-CMD RelayConfAns<br>Payload NOT = [0x]403F<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                             |                              |
| 11 | ED sends an Unconfirmed frame to the Relay using the default channel                             | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                                                |                              |
|    | Relay must send Unconfirmed frame to TCL                                                         | ← |   |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                              |                              |
|    | TCL sends Unconfirmed frame to Relay                                                             |   | ← |  | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 1 (500ms)<br>DefaultChIdx = a channel allowed for the region [2]<br>SecondChIdx = 1<br>SecondChDr = a DR NOT allowed for the region [2]<br>SecondChAckOffset = 0<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX | Invalid second channel DR    |
| 12 | Relay must send an Unconfirmed frame to the TCL                                                  |   | → |  | MAC-CMD RelayConfAns<br>Payload NOT = [0x]403F<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                             |                              |
| 13 | ED sends an Unconfirmed frame to the Relay using the default channel                             | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                                                |                              |
|    | Relay must send Unconfirmed frame to TCL                                                         | ← |   |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                              |                              |
|    | If the Relay belongs to a region that supports LR-FHSS [2], TCL sends Unconfirmed frame to Relay |   | ← |  | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 1 (500ms)<br>DefaultChIdx = a channel allowed for the region [2]<br>SecondChIdx = 1<br>SecondChDr = LR-FHSS DR [2]<br>SecondChAckOffset = 0<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX                      | Second channel DR = LR-FHSS  |

|    |                                                                                                             |   |   |  |                                                                                                                                                                                                                                                                                                                       |                            |
|----|-------------------------------------------------------------------------------------------------------------|---|---|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 14 | If the Relay belongs to a region that supports LR-FHSS [2], Relay must send an Unconfirmed frame to the TCL |   | → |  | MAC-CMD RelayConfAns<br>Payload NOT = [0x]403F<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                         |                            |
| 15 | ED sends an Unconfirmed frame to the Relay using the default channel                                        | → |   |  | <i>PHYPayload</i>                                                                                                                                                                                                                                                                                                     |                            |
|    | Relay must send Unconfirmed frame to TCL                                                                    | ← |   |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                          |                            |
|    | If the Relay belongs to a region that supports FSK [2], TCL sends Unconfirmed frame to Relay                |   | ← |  | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 1 (500ms)<br>DefaultChIdx = a channel allowed for the region [2]<br>SecondChIdx = 1<br>SecondChDr = FSK [2]<br>SecondChAckOffset = 0<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX                         | Second channel<br>DR = FSK |
| 16 | If the Relay belongs to a region that supports FSK [2], Relay must send an Unconfirmed frame to the TCL     |   | → |  | MAC-CMD RelayConfAns<br>Payload NOT = [0x]403F<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                         |                            |
| 17 | ED sends an Unconfirmed frame to the Relay using the default channel                                        | → |   |  | <i>PHYPayload</i>                                                                                                                                                                                                                                                                                                     |                            |
|    | Relay must send Unconfirmed frame to TCL                                                                    | ← |   |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                          |                            |
|    | TCL sends Unconfirmed frame to Relay                                                                        |   | ← |  | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 1 (500ms)<br>DefaultChIdx = a channel allowed for the region [2]<br>SecondChIdx = 1<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 6<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX | Invalid offset             |
| 18 | Relay must send an Unconfirmed frame to the TCL                                                             |   | → |  | MAC-CMD RelayConfAns<br>Payload NOT = [0x]403F<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                         |                            |
| 19 | ED sends an Unconfirmed frame to the Relay using the default channel                                        | → |   |  | <i>PHYPayload</i>                                                                                                                                                                                                                                                                                                     |                            |
|    | Relay must send Unconfirmed frame to TCL                                                                    | ← |   |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                          |                            |

|    |                                                                      |   |   |  |                                                                                                                                                                                                                                                                                                                                         |                        |
|----|----------------------------------------------------------------------|---|---|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
|    | TCL sends Unconfirmed frame to Relay                                 |   | ← |  | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 1 (500ms)<br>DefaultChIdx = a channel allowed for the region [2]<br>SecondChIdx = 1<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 0<br>SecondChFreq = a freq NOT allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX               | Invalid frequency      |
| 20 | Relay must send an Unconfirmed frame to the TCL                      |   | → |  | MAC-CMD RelayConfAns<br>Payload NOT = [0x]403F<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                                           |                        |
| 21 | ED sends an Unconfirmed frame to the Relay using the default channel | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                                                              |                        |
|    | Relay must send Unconfirmed frame to TCL                             | ← |   |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                                            |                        |
|    | TCL sends Unconfirmed frame to Relay                                 |   | ← |  | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = out of band value (invalid)<br>DefaultChIdx = a channel allowed for the region [2]<br>SecondChIdx = 1<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 0<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX | Invalid CADPeriodicity |
| 22 | Relay must send an Unconfirmed frame to the TCL                      |   | → |  | MAC-CMD RelayConfAns<br>Payload NOT = [0x]403F<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                                                                           |                        |
| 23 | ED sends an Unconfirmed frame to the Relay using the default channel | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                                                              |                        |
|    | Relay must send Unconfirmed frame to TCL                             | ← |   |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                                                            |                        |
|    | TCL sends Unconfirmed frame to Relay                                 |   | ← |  | MAC-CMD RelayConfReq<br>FPort = 0<br>StartStop = 1<br>CADPeriodicity = 0 (1s)<br>DefaultChIdx = a channel allowed for the region [2]<br>SecondChIdx = 0<br>SecondChDr = a DR allowed for the region [2]<br>SecondChAckOffset = 1 (200kHz)<br>SecondChFreq = a freq allowed for the region [2]<br>Payload = [0x]40XXXXXXXXXX             | Disable second channel |

|    |                                                                                              |   |   |   |                                                                                                                                                                                           |                      |
|----|----------------------------------------------------------------------------------------------|---|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 24 | Relay must send an Unconfirmed frame to the TCL                                              |   | → |   | MAC-CMD RelayConfAns<br>All Status ACK bits = 1<br>Payload = [0x]40XX<br><br>Note: This response may be piggy backed with the next ED uplink forward                                      |                      |
| 25 | ED sends an Unconfirmed frame to the Relay                                                   | → |   |   | PHYPayload                                                                                                                                                                                |                      |
|    | TCL sends Unconfirmed frame to ED directly                                                   |   |   | ← | MAC-CMD EndDeviceConfReq<br>FPort = 0<br>RelayModeActivation = 1<br>SecondChIdx, SecondChDR, SecondChAckOffset, SecondChFreq as set in RelayConfReq above<br>Payload = [0x]41XXXXXXXXXXXX |                      |
| 26 | ED sends the Relay an Unconfirmed frame using the default channel and NOT the second channel | → |   |   | PHYPayload includes<br>MAC-CMD EndDeviceConfAns<br>Payload = [0x]41XF                                                                                                                     | Default channel used |
|    | Relay must send an Unconfirmed frame to the TCL                                              |   | → |   | PHYPayload includes<br>MAC-CMD EndDeviceConfAns<br>Payload = [0x]41XF<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX |                      |

### 3.3.2 FilterListReq

- TCL sends the Relay a FilterListReq with
  - FilterListIdx = 0
  - FilterListAction = 2 (Filter)
  - FilterListLen = 0 bytes
  - FilterListEui = 0
- Relay must respond with the FilterListAns MAC command with bits set to 1 for bits 0-2
- TCL sends CP-CMD DutJoinReq to End-Device
- End-Device sends the Relay a Join-Request.
  - Relay must filter the Join-Request of the ED and must not forward it to the TCL
- TCL sends the Relay a FilterListReq with
  - FilterListIdx = 1
  - FilterListAction = 1 (Forward)
  - FilterListLen = 3 bytes
  - FilterListEui = leftmost 3 bytes of the concatenation of the JoinEUI and DevEUI of the End-Device
- Relay must respond with the FilterListAns MAC command with bits set to 1 for bits 0-2
- End-Device sends the Relay a Join-Request.
  - Relay must forward the ForwardUplinkReq (Join-Request) to the TCL with FPort = 226.
  - TCL sends a Join-Accept

- 536
  - Relay must send an empty uplink
- 537
  - TCL sends the UpdateUplinkListReq to the Relay
- 538
  - Relay responds with UpdateUplinkListAns successfully
- 539
- 540
  - TCL sends the Relay a FilterListReq with
- 541
  - FilterListIdx = 1
- 542
  - FilterListAction = 2 (Filter)
- 543
  - FilterListLen = 8 bytes
- 544
  - FilterListEui = leftmost 8 bytes of the concatenation of the JoinEUI and DevEUI of the
- 545
  - End-Device
- 546
  - Relay must respond with the FilterListAns MAC command with bits set to 1 for bits 0-2
- 547
  - TCL sends CP-CMD DutJoinReq to End-Device End-Device sends the Relay a Join-Request.
- 548
  - Relay must filter the Join-Request of the ED and must not forward it to the TCL
- 549
- 550
  - TCL sends the Relay a FilterListReq with
- 551
  - FilterListIdx = 2
- 552
  - FilterListAction = 1 (Forward)
- 553
  - FilterListLen = 15 bytes
- 554
  - FilterListEui = leftmost 15 bytes of the concatenation of the JoinEUI and DevEUI of the
- 555
  - End-Device
- 556
  - Relay must respond with the FilterListAns MAC command with bits set to 1 for bits 0-2
- 557
  - ED sends Join-Request to Relay
- 558
  - Relay must forward the ForwardUplinkReq (Join-Request) to the TCL with FPort = 226.
- 559
- 560
  - TCL sends the UpdateUplinkListReq to the Relay
- 561
  - Relay responds with UpdateUplinkListAns successfully
- 562
- 563
  - TCL sends the Relay a FilterListReq with
- 564
  - FilterListIdx = 3
- 565
  - FilterListAction = 2 (Filter)
- 566
  - FilterListLen = 16 bytes
- 567
  - FilterListEui = leftmost 16 bytes of the concatenation of the JoinEUI and DevEUI of the
- 568
  - End-Device
- 569
  - Relay must respond with the FilterListAns MAC command with bits set to 1 for bits 0-2
- 570
  - TCL sends CP-CMD DutJoinReq to End-Device End-Device sends the Relay a Join-Request.
- 571
  - Relay must filter the Join-Request of the ED and must not forward it to the TCL
- 572
- 573
  - TCL sends the Relay a FilterListReq with
- 574
  - FilterListIdx = 3
- 575
  - FilterListAction = 0 (No Rule)
- 576
  - FilterListLen = 0 bytes

- 577           ○ FilterListEui = 0
- 578       • Relay must respond with the FilterListAns MAC command with bits set to 1 for bits 0-2
- 579       • ED sends Join-Request to Relay
- 580           ○ Relay must forward the ForwardUplinkReq (Join-Request) to the TCL with FPort = 226.
- 581       • TCL sends the UpdateUplinkListReq to the Relay
- 582           ○ Relay responds with UpdateUplinkListAns successfully
- 583
- 584       • TCL sends the Relay a FilterListReq with
- 585           ○ FilterListIdx = 3
- 586           ○ FilterListAction = 3 (RFU)
- 587           ○ FilterListLen = 8 bytes
- 588           ○ FilterListEui = leftmost 8 bytes of the concatenation of the JoinEUI and DevEUI of the
- 589           End-Device
- 590       • Relay must respond with the FilterListAns MAC command with bit FilterListActionACK set to 0
- 591
- 592       • TCL sends the Relay a FilterListReq with
- 593           ○ FilterListIdx = 3
- 594           ○ FilterListAction = 2 (Filter)
- 595           ○ FilterListLen = 17
- 596           ○ FilterListEui = JoinEUI+DevEUI+0x01
- 597       • Relay must respond with the FilterListAns MAC command with FilterListLenACK set to bit 0
- 598
- 599       • TCL sends the Relay a FilterListReq with
- 600           ○ FilterListIdx = 3
- 601           ○ FilterListAction = 2 (Filter)
- 602           ○ FilterListLen = 0
- 603           ○ FilterListEui = empty
- 604       • Relay must respond with the FilterListAns MAC command with CombinedRulesACK set to bit
- 605       0
- 606       • TCL sends the Relay a FilterListReq with
- 607           ○ FilterListIdx = 0
- 608           ○ FilterListAction = 0 (No Rule)
- 609           ○ FilterListLen = 4 bytes
- 610           ○ FilterListEui = leftmost N bytes of the concatenation of the JoinEUI and DevEUI of the
- 611           End-Device
- 612       • Relay must respond with the FilterListAns MAC command with CombinedRulesACK set to bit
- 613       0
- 614
- 615       • TCL sends the Relay a FilterListReq with
- 616           ○ FilterListIdx = 3
- 617           ○ FilterListAction = 0 (No Rule)



- 618                   ○ FilterListLen = 5 bytes
- 619                   ○ FilterListEui = leftmost N bytes of the concatenation of the JoinEUI and DevEUI of the
- 620                   End-Device
- 621           • Relay must respond with the FilterListAns MAC command with CombinedRulesACK set to bit
- 622           0
- 623           • TCL sends CP-CMD DutJoinReq to End-Device End-Device sends the Relay a Join-Request.
- 624                   ○ Relay must forward the ForwardUplinkReq (Join-Request) to the TCL with FPort = 226.
- 625           • TCL sends the UpdateUplinkListReq to the Relay
- 626                   ○ Relay responds with UpdateUplinkListAns successfully
- 627

## 628       Test Procedure Frame Sequence Chart

629

|   |                                                    | ED<br><><br>R | R<><br>TCL | ED<br><><br>TCL | Frame                                                                                                                                                                                                                                                             |  |
|---|----------------------------------------------------|---------------|------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1 | ED sends an Unconfirmed frame to the Relay         | →             |            |                 | <i>PHYPayload</i>                                                                                                                                                                                                                                                 |  |
|   | Relay must send Unconfirmed frame to TCL           |               | →          |                 | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                      |  |
|   | TCL sends Unconfirmed frame to Relay               |               | ←          |                 | MAC-CMD FilterListReq<br>FPort = 0<br>FilterListIdx = 0<br>FilterListAction = 2 (Filter)<br>FilterListLen = 0 bytes<br>FilterListEui = 0<br>Payload = [0x]42XXXX                                                                                                  |  |
| 2 | Relay must send an Unconfirmed frame to the TCL    |               | →          |                 | MAC-CMD FilterListAns<br>Status Bits 0-2 = 1<br>Payload = [0x]42XX<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                 |  |
| 3 | ED sends an Unconfirmed frame to the Relay         | →             |            |                 | <i>PHYPayload</i>                                                                                                                                                                                                                                                 |  |
|   | Relay must send Unconfirmed frame to TCL           |               | →          |                 | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                      |  |
|   | TCL sends Unconfirmed frame to Relay               |               | ←          |                 | PHYPayload containing<br>CP-CMD DutJoinReq<br>FPort = 224<br>Payload = [0x]02<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226<br>Payload = [0x]variable                                                                                                     |  |
| 3 | ED sends an Unconfirmed frame to the Relay         | →             |            |                 | <i>PHYPayload containing Join-Request</i>                                                                                                                                                                                                                         |  |
|   | Relay must not forward the Join-Request to the TCL |               |            |                 |                                                                                                                                                                                                                                                                   |  |
| 4 | Relay must send an Unconfirmed frame to the TCL    |               | →          |                 |                                                                                                                                                                                                                                                                   |  |
|   | TCL sends Unconfirmed frame to Relay               |               | ←          |                 | MAC-CMD FilterListReq<br>FPort = 0<br>FilterListIdx = 1<br>FilterListAction = 1 (Forward)<br>FilterListLen = 3 bytes<br>FilterListEui = leftmost 3 bytes of the concatenation of the JoinEUI and DevEUI of the End-Device<br>Payload = [0x]42XXXX (var)3 bytes XX |  |

|    |                                                        |   |   |   |                                                                                                                                                                                                                                                                  |  |
|----|--------------------------------------------------------|---|---|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 5  | Relay sends an Unconfirmed frame to the TCL            |   | → |   | MAC-CMD FilterListAns<br>Status Bits 0-2 = 1<br>Payload = [0x]42XX<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                |  |
| 6  | ED sends Unconfirmed frame to Relay                    | → |   |   | <i>WORRelayJoin</i><br>and<br><i>PHYPayload</i> which includes the Join-Request                                                                                                                                                                                  |  |
|    | Relay sends Unconfirmed frame to the TCL               |   | → |   | ForwardUplinkReq (UplinkMetaData, UplinkFrequency, PHYPayload containing Join-Request)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXX                                                                                                                        |  |
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |   | ForwardDownlinkReq (PHYPayload containing Join-Accept)<br>FPort = 226<br>Payload = [0x]variable                                                                                                                                                                  |  |
| 7  | Relay must send an empty uplink to the TCL             |   | → |   |                                                                                                                                                                                                                                                                  |  |
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |   | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>RootWorSKey<br>DevAddr<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 3<br>UplinkLimitReloadRate = 63<br>WFCnt32 = 0<br>Payload = [0x]43 XX (26 octets)                                                             |  |
| 8  | Relay must send an Unconfirmed frame to the TCL        |   | → |   | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43                                                                                                                                                                                                                  |  |
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |   | MAC-CMD FilterListReq<br>FPort = 0<br>FilterListIdx = 1<br>FilterListAction = 2 (Filter)<br>FilterListLen = 8 bytes<br>FilterListEui = leftmost 8 bytes of the concatenation of the JoinEUI and DevEUI of the End-Device<br>Payload = [0x]42XXXX (var)8 bytes XX |  |
| 9  | Relay must send an Unconfirmed frame to the TCL        |   | → |   | MAC-CMD FilterListAns<br>Status Bits 0-2 = 1<br>Payload = [0x]42XX<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                |  |
| 10 | ED sends an Unconfirmed frame                          |   |   | → |                                                                                                                                                                                                                                                                  |  |
|    | TCL sends Unconfirmed frame to ED directly             |   |   | ← | CP-CMD DutJoinReq<br>FPort = 224<br>Payload = [0x]02                                                                                                                                                                                                             |  |
| 11 | ED sends Unconfirmed frame to Relay                    | → |   |   | <i>WORRelayJoin</i><br>and<br><i>PHYPayload</i> which includes the Join-Request                                                                                                                                                                                  |  |
|    | Relay must filter the ED and not forward it to the TCL |   | → |   |                                                                                                                                                                                                                                                                  |  |
| 12 | Relay must send an Unconfirmed frame to the TCL        |   | → |   |                                                                                                                                                                                                                                                                  |  |

|    |                                                        |   |   |   |                                                                                                                                                                                                                                                                      |  |
|----|--------------------------------------------------------|---|---|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |   | MAC-CMD FilterListReq<br>FPort = 0<br>FilterListIdx = 1<br>FilterListAction = 1 (Forward)<br>FilterListLen = 15 bytes<br>FilterListEui = leftmost 15 bytes of the concatenation of the JoinEUI and DevEUI of the End-Device<br>Payload = [0x]42XXXX (var)15 bytes XX |  |
| 13 | Relay sends an Unconfirmed frame to the TCL            |   | → |   | MAC-CMD FilterListAns<br>Status Bits 0-2 = 1<br>Payload = [0x]42XX<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                    |  |
| 14 | ED sends Unconfirmed frame to Relay                    | → |   |   | <i>WORRelayJoin and PHYPayload which includes the Join-Request</i>                                                                                                                                                                                                   |  |
|    | Relay sends Unconfirmed frame to the TCL               |   | → |   | ForwardUplinkReq (UplinkMetaData, UplinkFrequency, PHYPayload containing Join-Request)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXX                                                                                                                            |  |
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |   | ForwardDownlinkReq (PHYPayload containing Join-Accept)<br>FPort = 226<br>Payload = [0x]variable                                                                                                                                                                      |  |
| 15 | Relay must send an empty uplink to the TCL             |   | → |   |                                                                                                                                                                                                                                                                      |  |
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |   | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 3<br>UplinkLimitReloadRate = 63<br>RootWorSKey<br>DevAddr<br>WFCnt32 = 0<br>Payload = [0x]43 XX (26 octets)                                                                 |  |
| 16 | Relay must send an Unconfirmed frame to the TCL        |   | → |   | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                       |  |
| 17 | ED sends a Confirmed or Unconfirmed frame              | → |   |   |                                                                                                                                                                                                                                                                      |  |
|    | The TCL sends Unconfirmed frame to the ED              |   |   | ← | CP-CMD TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 20<br>Payload = [0x]0603<br>If ED uplink was a Confirmed frame, send ACK                                                                                                                               |  |
| 18 | ED sends a Confirmed or Unconfirmed frame to the Relay | → |   |   | <i>PHYPayload</i>                                                                                                                                                                                                                                                    |  |
|    | Relay must send Unconfirmed frame to TCL               |   | → |   | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXX                                                                                                                                                       |  |

|    |                                              |   |   |  |                                                                                                                                                                                                                                                                                                    |  |
|----|----------------------------------------------|---|---|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | PHYPayload includes<br>CP-CMD TxFramesCtrlReq<br>Frame type = Unconfirmed<br>Payload = [0x]0701<br>If ED uplink was a Confirmed frame, send ACK<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226                                                                                              |  |
| 19 | ED sends an Unconfirmed frame to the Relay   | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                         |  |
|    | Relay must send Unconfirmed frame to TCL     |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                       |  |
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | PHYPayload includes<br>If FCtrl ADR Bit = false, then perform this step, else skip to next downlink<br><br>CP-CMD ADRBitChangeReq-ON<br>FPort = 224<br>Payload = [0x]040<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226                                                                     |  |
| 20 | ED sends an Unconfirmed frame to the Relay   | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                         |  |
|    | Relay must send Unconfirmed frame to TCL     |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                       |  |
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | PHYPayload includes<br>MAC-CMD LinkADRReq<br>FPort = 0<br>DataRate = Max125kHzDR, refer <a href="#">[2]</a><br>ChMaskCntl:<br>DC = 0,<br>FC = 6<br>ChMask:<br>DC - Enable only default channels<br>FC = [0x]00FF<br>Payload = [0x]03XXXXXXXX<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226 |  |
| 21 | ED sends an Unconfirmed frame to the Relay   | → |   |  | PHYPayload includes<br>MAC-CMD LinkADRAns<br>Payload = [0x]0307                                                                                                                                                                                                                                    |  |
|    | Relay must send Unconfirmed frame to TCL     |   | → |  | PHYPayload includes<br>MAC-CMD LinkADRAns<br>Payload = [0x]0307<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                |  |

|    |                                                        |   |   |  |                                                                                                                                                                                                                                                                     |                 |
|----|--------------------------------------------------------|---|---|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 22 | ED sends an Unconfirmed frame to the Relay             | → |   |  | <i>PHYPayload</i>                                                                                                                                                                                                                                                   |                 |
|    | Relay must send Unconfirmed frame to TCL               |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                        |                 |
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |  | MAC-CMD FilterListReq<br>FPort = 0<br>FilterListIdx = 3<br>FilterListAction = 2 (Filter)<br>FilterListLen = 16 bytes<br>FilterListEui = leftmost 16 bytes of the concatenation of the JoinEUI and DevEUI of the End-Device<br>Payload = [0x]42XXXX (var)16 bytes XX |                 |
| 23 | Relay must send an Unconfirmed frame to the TCL        |   | → |  | MAC-CMD FilterListAns<br>Status Bits 0-2 = 1<br>Payload = [0x]42XX<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                   |                 |
| 24 | ED sends an Unconfirmed frame to the Relay             | → |   |  | <i>PHYPayload</i>                                                                                                                                                                                                                                                   |                 |
|    | Relay must send Unconfirmed frame to TCL               |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                        |                 |
|    | TCL sends Unconfirmed frame to ED                      |   | ← |  | PHYPayload containing<br>CP-CMD DutJoinReq<br>FPort = 224<br>Payload = [0x]02<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226<br>Payload = [0x]variable                                                                                                       |                 |
| 25 | ED sends Unconfirmed frame to Relay                    | → |   |  | <i>WORRelayJoin and PHYPayload which includes the Join-Request</i>                                                                                                                                                                                                  |                 |
|    | Relay must filter the ED and not forward it to the TCL |   | → |  |                                                                                                                                                                                                                                                                     |                 |
| 26 | Relay must send an Unconfirmed frame to the TCL        |   | → |  |                                                                                                                                                                                                                                                                     |                 |
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |  | MAC-CMD FilterListReq<br>FPort = 0<br>FilterListIdx = 3<br>FilterListAction = 0 (No Rule)<br>FilterListLen = 0 bytes<br>FilterListEui = 0<br>Payload = [0x]42XXXX                                                                                                   | Remove the rule |
| 27 | Relay must send an Unconfirmed frame to the TCL        |   | → |  | MAC-CMD FilterListAns<br>Status Bits 0-2 = 1<br>Payload = [0x]42XX                                                                                                                                                                                                  |                 |
| 28 | ED sends Unconfirmed frame to Relay                    | → |   |  | <i>WORRelayJoin and PHYPayload which includes the Join-Request</i>                                                                                                                                                                                                  |                 |

|    |                                                        |   |   |   |                                                                                                                                                                                                       |  |
|----|--------------------------------------------------------|---|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    | Relay sends Unconfirmed frame to the TCL               |   | → |   | ForwardUplinkReq (UplinkMetaData, UplinkFrequency, PHYPayload containing Join-Request)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXX                                                             |  |
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |   | ForwardDownlinkReq (PHYPayload containing Join-Accept)<br>FPort = 226<br>Payload = [0x]variable                                                                                                       |  |
| 29 | Relay must send an empty uplink to the TCL             |   | → |   |                                                                                                                                                                                                       |  |
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |   | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 3<br>UplinkLimitReloadRate = 63<br>RootWorSKey<br>DevAddr<br>WFCnt32 = 0<br>Payload = [0x]43 XX (26 octets)  |  |
| 30 | Relay must send an Unconfirmed frame to the TCL        |   | → |   | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                        |  |
| 31 | ED sends a Confirmed or Unconfirmed frame              |   |   | → |                                                                                                                                                                                                       |  |
|    | The TCL sends Unconfirmed frame to the ED directly     |   |   | ← | CP-CMD TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 20<br>Payload = [0x]0603<br>If ED uplink was a Confirmed frame, send ACK                                                                |  |
| 32 | ED sends a Confirmed or Unconfirmed frame to the Relay | → |   |   | PHYPayload                                                                                                                                                                                            |  |
|    | Relay must send Unconfirmed frame to TCL               |   | → |   | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXX                                                                                    |  |
|    | The TCL sends Unconfirmed frame to the Relay           |   | ← |   | PHYPayload includes<br>CP-CMD TxFramesCtrlReq<br>Frame type = Unconfirmed<br>Payload = [0x]0701<br>If ED uplink was a Confirmed frame, send ACK<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226 |  |
| 33 | ED sends an Unconfirmed frame to the Relay             | → |   |   | PHYPayload                                                                                                                                                                                            |  |
|    | Relay must send Unconfirmed frame to TCL               |   | → |   | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXX                                                                                    |  |

|    |                                              |   |   |  |                                                                                                                                                                                                                                                                                                                                                       |             |
|----|----------------------------------------------|---|---|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | <p>PHYPayload includes</p> <p>If FCtrl ADR Bit = false, then perform this step, else skip to next downlink</p> <p>CP-CMD AdrBitChangeReq-ON</p> <p>FPort = 224</p> <p>Payload = [0x]040</p> <p>ForwardDownlinkReq (PHYPayload)</p> <p>FPort = 226</p>                                                                                                 |             |
| 34 | ED sends an Unconfirmed frame to the Relay   | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                                                                            |             |
|    | Relay must send Unconfirmed frame to TCL     |   | → |  | <p>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)</p> <p>FPort = 226</p> <p>Payload = [0x]XXXXXXXXXXXX</p>                                                                                                                                                                                                                           |             |
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | <p>PHYPayload includes</p> <p>MAC-CMD LinkADRReq</p> <p>FPort = 0</p> <p>DataRate = Max125kHzDR, refer <a href="#">[2]</a></p> <p>ChMaskCntl:</p> <p>DC = 0,</p> <p>FC = 6</p> <p>ChMask:</p> <p>DC - Enable only default channels</p> <p>FC = [0x]00FF</p> <p>Payload = [0x]03XXXXXXXX</p> <p>ForwardDownlinkReq (PHYPayload)</p> <p>FPort = 226</p> |             |
| 35 | ED sends an Unconfirmed frame to the Relay   | → |   |  | <p>PHYPayload includes</p> <p>MAC-CMD LinkADRAns</p> <p>Payload = [0x]0307</p>                                                                                                                                                                                                                                                                        |             |
|    | Relay must send Unconfirmed frame to TCL     |   | → |  | <p>PHYPayload includes</p> <p>MAC-CMD LinkADRAns</p> <p>Payload = [0x]0307</p> <p>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)</p> <p>FPort = 226</p> <p>Payload = [0x]XXXXXXXXXXXX</p>                                                                                                                                            |             |
| 36 | ED sends Unconfirmed frame to the Relay      | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                                                                            |             |
|    | Relay must send Unconfirmed frame to TCL     |   | → |  | <p>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)</p> <p>FPort = 226</p> <p>Payload = [0x]XXXXXXXXXXXX</p>                                                                                                                                                                                                                           |             |
|    | TCL sends Unconfirmed frame to Relay         |   | ← |  | <p>MAC-CMD FilterListReq</p> <p>FPort = 0</p> <p>FilterListIdx = 3</p> <p>FilterListAction = 3 (RFU)</p> <p>FilterListLen = 8 bytes</p> <p>FilterListEui = leftmost 8 bytes of the concatenation of the JoinEUI and DevEUI of the End-Device</p> <p>Payload = [0x]42XXXX</p>                                                                          | Error tests |



|    |                                                 |   |   |  |                                                                                                                                                                                                                                                          |  |
|----|-------------------------------------------------|---|---|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 37 | Relay must send an Unconfirmed frame to the TCL |   | → |  | MAC-CMD FilterListAns<br>FilterListActionACK = 0<br>Payload = [0x]42XX                                                                                                                                                                                   |  |
|    | TCL sends Unconfirmed frame to Relay            |   | ← |  | MAC-CMD FilterListReq<br>FPort = 0<br>FilterListIdx = 3<br>FilterListAction = 2 (Filter)<br>FilterListLen = 17 bytes<br>FilterListEui = JoinEUI+DevEUI+0x01<br>Payload = [0x]42XXXX (var)17 bytes XX                                                     |  |
| 38 | Relay must send an Unconfirmed frame to the TCL |   | → |  | MAC-CMD FilterListAns<br>FilterListLenACK = 0<br>Payload = [0x]42XX                                                                                                                                                                                      |  |
|    | TCL sends Unconfirmed frame to Relay            |   | ← |  | MAC-CMD FilterListReq<br>FPort = 0<br>FilterListIdx = 3<br>FilterListAction = 2 (Filter)<br>FilterListLen = 0 bytes<br>FilterListEui = empty<br>Payload = [0x]42XXXX                                                                                     |  |
| 39 | Relay must send an Unconfirmed frame to the TCL |   | → |  | MAC-CMD FilterListAns<br>CombinedRulesACK = 0<br>Payload = [0x]42XX                                                                                                                                                                                      |  |
|    | TCL sends Unconfirmed frame to Relay            |   | ← |  | MAC-CMD FilterListReq<br>FPort = 0<br>FilterListIdx = 0<br>FilterListAction = 0<br>FilterListLen = 4 bytes<br>FilterListEui = leftmost 4 bytes of the concatenation of the JoinEUI and DevEUI of the End-Device<br>Payload = [0x]42XXXX (var)17 bytes XX |  |
| 40 | Relay must send an Unconfirmed frame to the TCL |   | → |  | MAC-CMD FilterListAns<br>CombinedRulesACK = 0<br>Payload = [0x]42XX                                                                                                                                                                                      |  |
|    | TCL sends Unconfirmed frame to Relay            |   | ← |  | MAC-CMD FilterListReq<br>FPort = 0<br>FilterListIdx = 3<br>FilterListAction = 0<br>FilterListLen = 5 bytes<br>FilterListEui = leftmost 5 bytes of the concatenation of the JoinEUI and DevEUI of the End-Device<br>Payload = [0x]42XXXX (var)17 bytes XX |  |
| 41 | Relay must send an Unconfirmed frame to the TCL |   | → |  | MAC-CMD FilterListAns<br>CombinedRulesACK = 0<br>Payload = [0x]42XX                                                                                                                                                                                      |  |
| 42 | ED sends an Unconfirmed frame to the Relay      | → |   |  | PHYPayload                                                                                                                                                                                                                                               |  |
|    | Relay must send Unconfirmed frame to TCL        |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXX                                                                                                                                            |  |
|    | TCL sends Unconfirmed frame                     |   | ← |  | PHYPayload containing<br>CP-CMD DutJoinReq<br>FPort = 224<br>Payload = [0x]02<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226<br>Payload = [0x]variable                                                                                            |  |

|    |                                                        |   |   |   |                                                                                                                                                                                                       |  |
|----|--------------------------------------------------------|---|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 43 | ED sends Unconfirmed frame to Relay                    | → |   |   | <i>WORRelayJoin</i><br>and<br><i>PHYPayload</i> which includes the <i>Join-Request</i>                                                                                                                |  |
|    | Relay sends Unconfirmed frame to the TCL               |   | → |   | ForwardUplinkReq (UplinkMetaData, UplinkFrequency, PHYPayload containing Join-Request)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXXXX                                                           |  |
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |   | ForwardDownlinkReq (PHYPayload containing Join-Accept)<br>FPort = 226<br>Payload = [0x]variable                                                                                                       |  |
| 44 | Relay must send an empty uplink to the TCL             |   | → |   |                                                                                                                                                                                                       |  |
|    | TCL sends Unconfirmed frame to Relay                   |   | ← |   | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 3<br>UplinkLimitReloadRate = 63<br>RootWorSKey<br>DevAddr<br>WFCnt32 = 0<br>Payload = [0x]43 XX (26 octets)  |  |
| 45 | Relay must send an Unconfirmed frame to the TCL        |   | → |   | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                        |  |
| 46 | ED sends a Confirmed or Unconfirmed frame              |   |   | → |                                                                                                                                                                                                       |  |
|    | The TCL sends Unconfirmed frame to the ED directly     |   |   | ← | CP-CMD TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 20<br>Payload = [0x]0603<br>If ED uplink was a Confirmed frame, send ACK                                                                |  |
| 47 | ED sends a Confirmed or Unconfirmed frame to the Relay | → |   |   | <i>PHYPayload</i>                                                                                                                                                                                     |  |
|    | Relay must send Unconfirmed frame to TCL               |   | → |   | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                          |  |
|    | The TCL sends Unconfirmed frame to the Relay           |   | ← |   | PHYPayload includes<br>CP-CMD TxFramesCtrlReq<br>Frame type = Unconfirmed<br>Payload = [0x]0701<br>If ED uplink was a Confirmed frame, send ACK<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226 |  |
| 48 | ED sends an Unconfirmed frame to the Relay             | → |   |   | <i>PHYPayload</i>                                                                                                                                                                                     |  |
|    | Relay must send Unconfirmed frame to TCL               |   | → |   | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                          |  |

|    |                                              |   |   |  |                                                                                                                                                                                                                                                                                                    |  |
|----|----------------------------------------------|---|---|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | PHYPayload includes<br>If FCtrl ADR Bit = false, then perform this step, else skip to next downlink<br><br>CP-CMD AdrBitChangeReq-ON<br>FPort = 224<br>Payload = [0x]040<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226                                                                     |  |
| 49 | ED sends an Unconfirmed frame to the Relay   | → |   |  | PHYPayload                                                                                                                                                                                                                                                                                         |  |
|    | Relay must send Unconfirmed frame to TCL     |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                                       |  |
|    | The TCL sends Unconfirmed frame to the Relay |   | ← |  | PHYPayload includes<br>MAC-CMD LinkADRReq<br>FPort = 0<br>DataRate = Max125kHzDR, refer <a href="#">[2]</a><br>ChMaskCntl:<br>DC = 0,<br>FC = 6<br>ChMask:<br>DC - Enable only default channels<br>FC = [0x]00FF<br>Payload = [0x]03XXXXXXXX<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226 |  |
| 50 | ED sends an Unconfirmed frame to the Relay   | → |   |  | PHYPayload includes<br>MAC-CMD LinkADRAns<br>Payload = [0x]0307                                                                                                                                                                                                                                    |  |
|    | Relay must send Unconfirmed frame to TCL     |   | → |  | PHYPayload includes<br>MAC-CMD LinkADRAns<br>Payload = [0x]0307<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                |  |

630

631

### 632 3.3.3 UpdateUplinkListReq

- 633     • TCL sends the Relay an UpdateUplinkListReq with
- 634         ○ UpdateListIdx = 1
- 635         ○ UpdateLimitBucketSize = 0
- 636         ○ UpdateLimitReloadRate = 0
- 637             ▪ DevAddr of the End-Device
- 638         ○ Valid WFCnt32

- 639                   ○ RootWorSKey = key associated to the DevAddr of the End-Device
- 640           • Relay must respond with the UpdateUplinkListAns MAC command
- 641           • End-Device sends an uplink to the Relay
- 642                   ○ Relay must NOT forward the uplink to the TCL as it has been disabled
- 643
- 644           • TCL sends the Relay a UpdateUplinkListReq with
- 645                   ○ UpdateListIdx = 1
- 646                   ○ UpdateLimitBucketSize = 1
- 647                   ○ UpdateLimitReloadRate = 63
- 648                   ○ DevAddr of the End-Device
- 649                   ○ Valid WFCnt32
- 650                   ○ RootWorSKey = key associated to the DevAddr of the End-Device
- 651           • Relay must respond with the UpdateUplinkListAns MAC command
- 652           • End-Device sends an uplink to the Relay
- 653                   ○ Relay must forward the uplink to the TCL
- 654           • End-Device sends another uplink to the Relay
- 655                   ○ Relay must forward the uplink to the TCL
- 656
- 657           • TCL sends the Relay an UpdateUplinkListReq with
- 658                   ○ UpdateListIdx = 1
- 659                   ○ UpdateLimitBucketSize = 0
- 660                   ○ UpdateLimitReloadRate = 1
- 661                   ○ DevAddr of the End-Device
- 662                   ○ WFCnt32 = large INVALID value
- 663                   ○ RootWorSKey = key associated to the DevAddr of the End-Device
- 664           • Relay must respond with the UpdateUplinkListAns MAC command
- 665           • End-Device sends an uplink to the Relay
- 666                   ○ Relay must NOT forward the uplink to the TCL as it thinks that it is an unknown End-
- 667                   Device. Relay must instead send a NotifyNewEndDeviceReq to the TCL.
- 668           • TCL sends UpdateUplinkListReq to the Relay
- 669                   ○ Relay must respond with UpdateUplinkListAns successfully
- 670           • End-Device sends an uplink to the Relay
- 671                   ○ Relay forwards the uplink to the TCL
- 672
- 673

## Test Procedure Frame Sequence Chart

|   |                                                                      | ED<br><><br>R<br>→ | R<><br>TCL | ED<br><><br>TCL | Frame                                                                                                                                                                                                                                                                    |  |
|---|----------------------------------------------------------------------|--------------------|------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1 | ED sends an Unconfirmed frame to the Relay                           | →                  |            |                 | <i>PHYPayload</i>                                                                                                                                                                                                                                                        |  |
|   | Relay must send Unconfirmed frame to TCL                             |                    | →          |                 | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                             |  |
|   | TCL sends Unconfirmed frame to Relay                                 |                    | ←          |                 | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 0<br>UplinkLimitReloadRate = 0<br>DevAddr<br>WFCnt32 = next expected value<br>RootWorSKey = key associated to the DevAddr of the End-Device<br>Payload = [0x]43 XX (26 octets)  |  |
| 2 | Relay must send an Unconfirmed frame to the TCL                      |                    | →          |                 | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                           |  |
| 3 | ED sends the Relay an Unconfirmed frame                              | →                  |            |                 | <i>PHYPayload</i>                                                                                                                                                                                                                                                        |  |
| 4 | Relay must not forward the uplink to the TCL as it has been disabled |                    |            |                 |                                                                                                                                                                                                                                                                          |  |
| 5 | Relay must send an Unconfirmed frame to the TCL                      |                    | →          |                 |                                                                                                                                                                                                                                                                          |  |
|   | TCL sends Unconfirmed frame to Relay                                 |                    | ←          |                 | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 1<br>UplinkLimitReloadRate = 63<br>DevAddr<br>WFCnt32 = next expected value<br>RootWorSKey = key associated to the DevAddr of the End-Device<br>Payload = [0x]43 XX (26 octets) |  |
| 6 | Relay must send an Unconfirmed frame to the TCL                      |                    | →          |                 | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                           |  |
| 7 | ED sends an Unconfirmed frame to the Relay                           | →                  |            |                 | <i>PHYPayload</i>                                                                                                                                                                                                                                                        |  |
|   | Relay must send Unconfirmed frame to TCL                             |                    | →          |                 | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                             |  |
| 8 | ED sends another Unconfirmed frame to the Relay                      | →                  |            |                 | <i>PHYPayload</i>                                                                                                                                                                                                                                                        |  |

|    |                                                 |   |   |  |                                                                                                                                                                                                                                                                                  |  |
|----|-------------------------------------------------|---|---|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    | Relay must send Unconfirmed frame to TCL        |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                     |  |
| 9  | Relay must send an Unconfirmed frame to the TCL |   | → |  |                                                                                                                                                                                                                                                                                  |  |
|    | TCL sends Unconfirmed frame to Relay            |   | ← |  | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 0<br>UplinkLimitReloadRate = 1<br>DevAddr<br>WFCnt32 = large value which is INVALID<br>RootWorSKey = key associated to the DevAddr of the End-Device<br>Payload = [0x]43 XX (26 octets) |  |
| 10 | Relay must send an Unconfirmed frame to the TCL |   | → |  | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43                                                                                                                                                                                                                                  |  |
| 11 | ED sends another Unconfirmed frame to the Relay | → |   |  | PHYPayload                                                                                                                                                                                                                                                                       |  |
|    | Relay must send an Unconfirmed frame to the TCL |   | → |  | MAC-CMD NotifyNewEndDeviceReq<br>Payload = [0x]46XXXXXXXXXXXX                                                                                                                                                                                                                    |  |
|    | TCL sends Unconfirmed frame to Relay            |   | ← |  | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>RootWorSKey<br>DevAddr<br>WFCnt32 = next expected value<br>UplinkLimitBucketSize = 2<br>UplinkLimitReloadRate = 63<br>Payload = [0x]43 XX (26 octets)                                                           |  |
| 12 | Relay must send an Unconfirmed frame to the TCL |   | → |  | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                   |  |
| 13 | ED sends another Unconfirmed frame to the Relay | → |   |  | PHYPayload                                                                                                                                                                                                                                                                       |  |
|    | Relay must send Unconfirmed frame to TCL        |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                                     |  |

676

677

678 The following tests must be displayed as a separate test in the test tool as it takes 1 hour to execute.

679 • TCL sends the Relay an UpdateUplinkListReq with

- 680 ○ UpdateListIdx = 1
- 681 ○ UpdateLimitBucketSize = 0
- 682 ○ UpdateLimitReloadRate = 1
- 683 ○ DevAddr of the End-Device
- 684 ○ Valid WFCnt32

- 685                   ○ RootWorSKey = key associated to the DevAddr of the End-Device
- 686           • Relay must respond with the UpdateUplinkListAns MAC command
- 687           • End-Device sends an uplink to the Relay
- 688                   ○ Relay must forward the uplink to the TCL
- 689           • End-Device sends another uplink to the Relay
- 690                   ○ Relay must ignore the uplink as the limit has reached (which was set to 1 token every
- 691                   hour)
- 692           • ED waits for 1 hour and sends another uplink to the Relay
- 693                   ○ Relay must forward the uplink to the TCL
- 694
- 695

|   | Description                                                                                            | ED<br><><br>R | R<><br>TCL | ED<br><><br>TCL | Frame                                                                                                                                                                                                                                                                   |  |
|---|--------------------------------------------------------------------------------------------------------|---------------|------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1 | Relay must send an Unconfirmed frame to the TCL                                                        |               | →          |                 |                                                                                                                                                                                                                                                                         |  |
|   | TCL sends Unconfirmed frame to Relay                                                                   |               | ←          |                 | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 0<br>UplinkLimitReloadRate = 1<br>DevAddr<br>WFCnt32 = next expected value<br>RootWorSKey = key associated to the DevAddr of the End-Device<br>Payload = [0x]43 XX (26 octets) |  |
| 2 | Relay must send an Unconfirmed frame to the TCL                                                        |               | →          |                 | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                          |  |
| 3 | ED sends an Unconfirmed frame to the Relay                                                             | →             |            |                 | PHYPayload                                                                                                                                                                                                                                                              |  |
|   | Relay must send Unconfirmed frame to TCL                                                               |               | →          |                 | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                            |  |
| 4 | ED sends an Unconfirmed frame to the Relay                                                             | →             |            |                 | PHYPayload                                                                                                                                                                                                                                                              |  |
|   | Relay must ignore the uplink as the limit has been reached (1 token every hour)<br><br>Wait for 1 hour |               | →          |                 |                                                                                                                                                                                                                                                                         |  |
| 5 | ED sends an Unconfirmed frame to the Relay                                                             | →             |            |                 | PHYPayload                                                                                                                                                                                                                                                              |  |
|   | Relay must send Unconfirmed frame to TCL                                                               |               | →          |                 | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                                                                                                                            |  |

696

### 697 3.3.4 CtrlUplinkListReq

- 698 • TCL sends the Relay a CtrlUplinkListReq with
  - 699 ○ UpdateListIdx = 1
  - 700 ○ CtrlUplinkAction = 1
- 701 • Relay must respond with the CtrlUplinkListAns MAC command with UplinkListIdxACK set to 1
- 702 and WFCnt32 = next valid value expected by the Relay
- 703 • End-Device (which was removed from the trusted list) sends another uplink to the Relay
  - 704 ○ Relay must send a NotifyNewEndDeviceReq to the TCL.
- 705 • TCL sends the Relay a UpdateUplinkListReq with



- 706
  - UpdateListIdx = 1
  - 707○ UpdateLimitBucketSize = 0
  - 708○ UpdateLimitReloadRate = 5
  - 709○ DevAddr of the End-Device
  - 710○ Valid WFCnt32
  - 711○ RootWorSKey = key associated to the DevAddr of the End-Device
- 712• Relay must respond with the UpdateUplinkListAns MAC command
- 713
- 714• TCL sends the Relay a CtrlUplinkListReq with
  - 715○ UpdateListIdx = 1
  - 716○ CtrlUplinkAction = 0
- 717• Relay must respond with the CtrlUplinkListAns MAC command with UplinkListIdxACK set to 1
- 718and WFCnt32 = next valid value expected by the Relay
- 719• End-Device (which was added to the trusted list) sends another uplink to the Relay
  - 720○ Relay must forward the uplink to the TCL.
- 721
- 722• TCL sends the Relay a CtrlUplinkListReq with
  - 723○ UpdateListIdx = 0
  - 724○ CtrlUplinkAction = 1
- 725• Relay must respond with the CtrlUplinkListAns MAC command with an Error
- 726
- 727• TCL sends the Relay a CtrlUplinkListReq with
  - 728○ UpdateListIdx = 4
  - 729○ CtrlUplinkAction = 0
- 730• Relay must respond with the CtrlUplinkListAns MAC command with UplinkListIdxACK set to 0
- 731

## 732 Test Procedure Frame Sequence Chart

733

|   |                                                                             | ED<br><><br>R | R<><br>TCL | ED<br><><br>TCL | Frame                                                                                                                                                                                                                                                                                     |  |
|---|-----------------------------------------------------------------------------|---------------|------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1 | Relay must send an Unconfirmed frame to the TCL                             |               | →          |                 |                                                                                                                                                                                                                                                                                           |  |
|   | TCL sends Unconfirmed frame to Relay                                        |               | ←          |                 | MAC-CMD CtrlUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>CtrlUplinkAction = 1<br>Payload = [0x]44XX                                                                                                                                                                                 |  |
| 2 | Relay must send an Unconfirmed frame to the TCL                             |               | →          |                 | MAC-CMD CtrlUplinkListAns<br>UplinkListIdxACK = 1<br>WFCnt32 = next valid value expected by the Relay<br>Payload = [0x]44XXXXXXXXXX                                                                                                                                                       |  |
| 3 | ED sends an Unconfirmed frame to the Relay                                  | →             |            |                 | PHYPayload                                                                                                                                                                                                                                                                                |  |
|   | Relay must not forward the ED uplink to the TCL as the ED has been disabled |               |            |                 |                                                                                                                                                                                                                                                                                           |  |
| 4 | Relay must send an Unconfirmed frame to the TCL                             |               | →          |                 | MAC-CMD NotifyNewEndDeviceReq<br>Payload = [0x]46XXXXXXXXXXXXXX                                                                                                                                                                                                                           |  |
|   | TCL sends Unconfirmed frame to Relay                                        |               | ←          |                 | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UpdateLimitBucketSize = 0<br>UpdateLimitReloadRate = 5<br>DevAddr of the End-Device<br>WFCnt32 = next expected value<br>RootWorSKey = key associated to the DevAddr of the End-Device<br>Payload = [0x]43 XX (26 octets) |  |
| 5 | Relay must send an Unconfirmed frame to the TCL                             |               | →          |                 | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                            |  |
|   | TCL sends Unconfirmed frame to Relay                                        |               | ←          |                 | MAC-CMD CtrlUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>CtrlUplinkAction = 0<br>Payload = [0x]44XX                                                                                                                                                                                 |  |
| 6 | Relay must send an Unconfirmed frame to the TCL                             |               | →          |                 | MAC-CMD CtrlUplinkListAns<br>UplinkListIdxACK = 1<br>WFCnt32 = next valid value expected by the Relay<br>Payload = [0x]44XXXXXXXXXX<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                        |  |
| 7 | ED sends an Unconfirmed frame to the Relay                                  | →             |            |                 | PHYPayload                                                                                                                                                                                                                                                                                |  |
|   | Relay must send Unconfirmed frame to TCL                                    |               | →          |                 | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXX                                                                                                                                                                            |  |

|    |                                                 |   |   |  |                                                                                                                                                                |  |
|----|-------------------------------------------------|---|---|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    | TCL sends Unconfirmed frame to Relay            |   | ← |  | MAC-CMD CtrlUplinkListReq<br>FPort = 0<br>UpdateListIdx = 0<br>CtrlUplinkAction = 1<br>Payload = [0x]44XX                                                      |  |
| 8  | Relay must send an Unconfirmed frame to the TCL |   | → |  | MAC-CMD CtrlUplinkListAns<br>UplinkListIdxACK = 0<br>Payload = [0x]44XXXXXXXXXX<br><br>Note: This response may be piggy backed with the next ED uplink forward |  |
|    | TCL sends Unconfirmed frame to Relay            |   | ← |  | MAC-CMD CtrlUplinkListReq<br>FPort = 0<br>UpdateListIdx = 4<br>CtrlUplinkAction = 0<br>Payload = [0x]44XX                                                      |  |
| 9  | Relay must send an Unconfirmed frame to the TCL |   | → |  | MAC-CMD CtrlUplinkListAns<br>UplinkListIdxACK = 0<br>Payload = [0x]44XXXXXXXXXX<br><br>Note: This response may be piggy backed with the next ED uplink forward |  |
| 10 | ED sends an Unconfirmed frame to the Relay      | → |   |  | PHYPayload                                                                                                                                                     |  |
|    | Relay must send Unconfirmed frame to TCL        |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXX                                                   |  |

734

735

### 736 3.3.5 ConfigureFwdLimitReq

737 Note: This test may be performed as part of Pre-testing by the manufacturer and the test log may be  
738 submitted to the ATH before certification testing. In this case, the ATH need not repeat this test during  
739 official certification testing.

740

- 741 • TCL sends the Relay a ConfigureFwdLimitReq with
  - 742 ○ ResetLimitCounter = 1 (reload rate)
  - 743 ○ JoinReqReloadRate = 1
  - 744 ○ NotifyReloadRate = 1
  - 745 ○ GlobalUplinkReloadRate = 20
  - 746 ○ OverallReloadRate = 40
  - 747 ○ FwdLimitLoadCapacityPL = 0xFF
- 748 • Relay must respond with the ConfigureFwdLimitAns MAC command with no error
- 749
- 750 • TCL sends CP-CMD DutJoinReq to End-Device
- 751 • ED sends the Relay a Join-Request.
  - 752 ○ Relay must forward the ForwardUplinkReq (Join-Request) to the TCL with FPort = 226.
- 753 • TCL does not respond with Join-Accept

- 754       • ED sends the Relay another Join-Request  
755       • Relay does not forward the Join-Request to the TCL as the limit is reached  
756

757 **Notify Limit tests**

- 758       • TCL sends the Relay a ConfigureFwdLimitReq with  
759           ○ ResetLimitCounter = 1 (reload rate)  
760           ○ JoinReqReloadRate = 1  
761           ○ NotifyReloadRate = 1  
762           ○ GlobalUplinkReloadRate = 20  
763           ○ OverallReloadRate = 40  
764           ○ FwdLimitLoadCapacityPL = 0xFF  
765       • Relay must respond with the ConfigureFwdLimitAns MAC command  
766       • ED sends the Relay a Join-Request.  
767           ○ Relay must forward the ForwardUplinkReq (Join-Request) to the TCL with FPort = 226.  
768       • TCL sends the Join-Accept  
769       • TCL sends the Relay a UpdateUplinkListReq command with an invalid WFCnt32 value  
770       • Relay must not forward the uplink from the ED to the TCL  
771       • Relay must send the TCL a NotifyNewEndDeviceReq command  
772       • TCL replies with a UpdateUplinkListReq command with a valid WFCnt32 value  
773       • TCL changes ED TxPeriodicity again to 20 seconds  
774       • TCL sends the Relay another UpdateUplinkListReq command with an invalid WFCnt32 value  
775       • Relay must not send the TCL a NotifyNewEndDeviceReq command as the limit for Notify  
776        commands has been reached  
777

778 **Uplink Limit Tests**

- 779       • TCL sends the Relay a ConfigureFwdLimitReq with  
780           ○ ResetLimitCounter = 1 (reload rate)  
781           ○ JoinReqReloadRate = 10  
782           ○ NotifyReloadRate = 10  
783           ○ GlobalUplinkReloadRate = 1  
784           ○ OverallReloadRate = 40  
785           ○ FwdLimitLoadCapacityPL = 0xFF  
786       • Relay must respond with the ConfigureFwdLimitAns MAC command  
787       • End-Device sends the Relay the uplink frame.  
788       • Relay sends NotifyNewEndDeviceReq command again to TCL  
789       • TCL sends the Relay the UpdateUplinkListReq command  
790       • Relay must respond with UpdateUplinkListAns command  
791       • ED sends uplink to the Relay  
792           ○ Relay must send the ForwardUplinkReq (Uplink) to the TCL with FPort = 226  
793       • End-Device sends the Relay another uplink frame

- 794                   ○ Relay must not forward the uplink to the TCL as the limit has been reached
- 795
- 796           • TCL sends the Relay a ConfigureFwdLimitReq with
- 797                   ○ ResetLimitCounter = 1 (reload rate)
- 798                   ○ JoinReqReloadRate = 5
- 799                   ○ NotifyReloadRate = 5
- 800                   ○ GlobalUplinkReloadRate = 5
- 801                   ○ OverallReloadRate = 1
- 802                   ○ FwdLimitLoadCapacityPL = 0xFF
- 803           • Relay must respond with the ConfigureFwdLimitAns MAC command
- 804           • ED sends the Relay an uplink frame
- 805           • Relay must forward the uplink to TCL
- 806           • ED sends the Relay another uplink frame.
- 807                   ○ Relay must not forward the uplink to the TCL as the overall limit is reached
- 808
- 809           • TCL sends the Relay the MAC-CMD ConfigureFwdLimitReq with
- 810                   ResetLimitCounter = 1 (reload rate)
- 811                   JoinReqReloadRate = 127
- 812                   NotifyReloadRate = 127
- 813                   GlobalUplinkReloadRate = 127
- 814                   OverallReloadRate = 127
- 815                   FwdLimitLoadCapacityPL = 0xFF
- 816
- 817                   ○ Relay must respond with ConfigureFwdLimitAns
- 818           • ED sends the Relay an uplink
- 819                   ○ Relay must forward the uplink successfully to the TCL
- 820           • Enable the Duty Cycle with CP-CMD RegionalDutyCycleCtrlReq
- 821

## 822       Test Procedure Frame Sequence Chart

823

|   |                                                             | ED<br><><br>R | R<><br>TCL | ED<br><><br>TCL | Frame                                                                                                                                                                                                                                                       |  |
|---|-------------------------------------------------------------|---------------|------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1 | Relay must send an Unconfirmed frame to the TCL             |               | →          |                 |                                                                                                                                                                                                                                                             |  |
|   | TCL sends Unconfirmed frame to Relay                        |               | ←          |                 | MAC-CMD ConfigureFwdLimitReq<br>FPort = 0<br>ResetLimitCounter = 1 (reload rate)<br>JoinReqReloadRate = 1<br>NotifyReloadRate = 1<br>GlobalUplinkReloadRate = 20<br>OverallReloadRate = 40<br>FwdLimitLoadCapacityPL = 0xFF<br>Payload = [0x]45XXXXXXXXXXXX |  |
| 2 | Relay must send an Unconfirmed frame to the TCL             |               | →          |                 | MAC-CMD ConfigureFwdLimitAns<br>Payload = [0x]45<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                             |  |
| 3 | ED sends Unconfirmed frame to Relay                         | →             |            |                 | <i>PHYPayload</i>                                                                                                                                                                                                                                           |  |
|   | Relay sends Unconfirmed frame to the TCL                    |               | →          |                 | ForwardUplinkReq (UplinkMetaData, UplinkFrequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXXXX                                                                                                                                         |  |
|   | TCL sends Unconfirmed frame                                 |               |            | ←               | ForwardDownlinkReq (PHYPayload containing CP-CMD DutJoinReq)<br>FPort = 226<br>Payload = [0x]variable                                                                                                                                                       |  |
| 4 | ED sends Unconfirmed frame to Relay                         | →             |            |                 | <i>PHYPayload which includes the Join-Request</i>                                                                                                                                                                                                           |  |
|   | Relay sends Unconfirmed frame to the TCL                    |               | →          |                 | ForwardUplinkReq (UplinkMetaData, UplinkFrequency, PHYPayload containing Join-Request)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXXXX                                                                                                                 |  |
|   | TCL does NOT respond with Join-Accept                       |               |            |                 |                                                                                                                                                                                                                                                             |  |
| 5 | ED sends Unconfirmed frame to Relay                         | →             |            |                 | <i>PHYPayload which includes the Join-Request</i>                                                                                                                                                                                                           |  |
|   | Relay must not forward the Join-Request as limit is reached |               |            |                 |                                                                                                                                                                                                                                                             |  |
| 6 | Relay must send an Unconfirmed frame to the TCL             |               | →          |                 |                                                                                                                                                                                                                                                             |  |
|   | TCL sends Unconfirmed frame to Relay                        |               | ←          |                 | MAC-CMD ConfigureFwdLimitReq<br>FPort = 0<br>ResetLimitCounter = 1 (reload rate)<br>JoinReqReloadRate = 1<br>NotifyReloadRate = 1<br>GlobalUplinkReloadRate = 20<br>OverallReloadRate = 40<br>FwdLimitLoadCapacityPL = 0xFF<br>Payload = [0x]45XXXXXXXXXXXX |  |

|                                          |                                                 |   |   |   |                                                                                                                                                                                                                                                                                  |                                                |
|------------------------------------------|-------------------------------------------------|---|---|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| 7                                        | Relay must send an Unconfirmed frame to the TCL |   | → |   | MAC-CMD ConfigureFwdLimitAns<br>Payload = [0x]45<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                  |                                                |
| 8                                        | ED sends Unconfirmed frame to Relay             | → |   |   | PHYPayload which includes the Join-Request                                                                                                                                                                                                                                       |                                                |
|                                          | Relay sends Unconfirmed frame to the TCL        |   | → |   | ForwardUplinkReq (UplinkMetaData, UplinkFrequency, PHYPayload containing Join-Request)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXXXX                                                                                                                                      |                                                |
|                                          | TCL sends Unconfirmed frame to Relay            |   | ← |   | ForwardDownlinkReq (PHYPayload containing Join-Accept)<br>FPort = 226<br>Payload = [0x]variable                                                                                                                                                                                  |                                                |
| 9                                        | Relay must send an Unconfirmed frame to the TCL |   | → |   |                                                                                                                                                                                                                                                                                  | TCL does not respond with UpdateUplinkListReq. |
| Either Step 10 or 11 could occur earlier |                                                 |   |   |   |                                                                                                                                                                                                                                                                                  |                                                |
| 10                                       | ED must send Confirmed or Unconfirmed frame     |   |   | → |                                                                                                                                                                                                                                                                                  |                                                |
|                                          | TCL sends Unconfirmed frame                     |   |   | ← | If ED sent Confirmed uplink frame, TCL must Acknowledge<br><br>CP-CMD TxPeriodicityChangeReq<br>FPort = 224<br>Periodicity = 20<br>Payload = [0x]0603                                                                                                                            |                                                |
| 11                                       | Relay must send an Unconfirmed frame to the TCL |   | → |   | MAC-CMD NotifyNewEndDeviceReq<br>Payload = [0x]46XXXXXXXXXXXX                                                                                                                                                                                                                    | Testing for Notify limit                       |
|                                          | TCL sends Unconfirmed frame to Relay            |   | ← |   | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>UpdateListIdx = 1<br>UplinkLimitBucketSize = 0<br>UplinkLimitReloadRate = 1<br>DevAddr<br>WFCnt32 = large value which is INVALID<br>RootWorSKey = key associated to the DevAddr of the End-Device<br>Payload = [0x]43 XX (26 octets) |                                                |
| 12                                       | Relay must send an Unconfirmed frame to the TCL |   | → |   | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                                   |                                                |
| 13                                       | ED sends another Unconfirmed frame to the Relay | → |   |   | PHYPayload                                                                                                                                                                                                                                                                       |                                                |

|    |                                                                       |   |   |  |                                                                                                                                                                                                                                                            |                       |
|----|-----------------------------------------------------------------------|---|---|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    | Relay does not send the NotifyNewEndDeviceReq as the limit is reached |   |   |  |                                                                                                                                                                                                                                                            | Notify limit reached  |
| 14 | Relay must send an Unconfirmed frame to the TCL                       |   | → |  |                                                                                                                                                                                                                                                            |                       |
|    | TCL sends Unconfirmed frame to Relay                                  |   | ← |  | MAC-CMD ConfigureFwdLimitReq<br>FPort = 0<br>ResetLimitCounter = 1 (reload rate)<br>JoinReqReloadRate = 10<br>NotifyReloadRate = 10<br>GlobalUplinkReloadRate = 1<br>OverallReloadRate = 40<br>FwdLimitLoadCapacityPL = 0xFF<br>Payload = [0x]45XXXXXXXXXX |                       |
| 15 | Relay must send an Unconfirmed frame to the TCL                       |   | → |  | MAC-CMD ConfigureFwdLimitAns<br>Payload = [0x]45<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                            |                       |
| 16 | ED sends an Unconfirmed frame to the Relay                            | → |   |  | PHYPayload                                                                                                                                                                                                                                                 | Checking uplink limit |
|    | Relay must send Unconfirmed frame to TCL                              |   | → |  | MAC-CMD NotifyNewEndDeviceReq<br>Payload = [0x]46XXXXXXXXXXXXXX                                                                                                                                                                                            |                       |
|    | TCL sends Unconfirmed frame to Relay                                  |   | ← |  | MAC-CMD UpdateUplinkListReq<br>FPort = 0<br>RootWorSKey<br>UpdateListIdx = 1<br>DevAddr<br>WFCnt32 = next expected value<br>UplinkLimitBucketSize = 2<br>UplinkLimitReloadRate = 63<br>Payload = [0x]43 XX (26 octets)                                     |                       |
| 17 | Relay must send an Unconfirmed frame to the TCL                       |   | → |  | MAC-CMD UpdateUplinkListAns<br>Payload = [0x]43<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                             |                       |
| 18 | ED sends an Unconfirmed frame to the Relay                            | → |   |  | PHYPayload                                                                                                                                                                                                                                                 |                       |
|    | Relay sends Unconfirmed frame to the TCL                              |   | → |  | ForwardUplinkReq (UplinkMetaData, UplinkFrequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXX                                                                                                                                          |                       |
| 19 | ED sends Unconfirmed frame to Relay                                   | → |   |  | PHYPayload                                                                                                                                                                                                                                                 |                       |
|    | Relay must not forward the uplink as the limit is reached             |   |   |  |                                                                                                                                                                                                                                                            | Uplink limit reached  |
| 20 | Relay must send Unconfirmed frame to TCL                              |   |   |  |                                                                                                                                                                                                                                                            |                       |



|    |                                                                       |   |   |  |                                                                                                                                                                                                                                                                 |  |
|----|-----------------------------------------------------------------------|---|---|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    | TCL sends Unconfirmed frame to Relay                                  |   | ← |  | MAC-CMD ConfigureFwdLimitReq<br>FPort = 0<br>ResetLimitCounter = 1 (reload rate)<br>JoinReqReloadRate = 5<br>NotifyReloadRate = 5<br>GlobalUplinkReloadRate = 5<br>OverallReloadRate = 1<br>FwdLimitLoadCapacityPL = 0xFF<br>Payload = [0x]45XXXXXXXXXX         |  |
| 21 | Relay must send an Unconfirmed frame to the TCL                       |   | → |  | MAC-CMD ConfigureFwdLimitAns<br>Payload = [0x]45<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                 |  |
| 22 | ED sends Unconfirmed frame to Relay                                   | → |   |  | PHYPayload                                                                                                                                                                                                                                                      |  |
|    | Relay sends Unconfirmed frame to the TCL                              |   | → |  | ForwardUplinkReq (UplinkMetaData, UplinkFrequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXX                                                                                                                                               |  |
| 23 | ED sends Unconfirmed frame to Relay                                   | → |   |  | PHYPayload                                                                                                                                                                                                                                                      |  |
|    | Relay must not forward any uplink from ED as overall limit is reached |   |   |  |                                                                                                                                                                                                                                                                 |  |
| 24 | Relay must send an Unconfirmed frame to the TCL                       |   | → |  |                                                                                                                                                                                                                                                                 |  |
|    | TCL sends Unconfirmed frame to Relay                                  |   | ← |  | MAC-CMD ConfigureFwdLimitReq<br>FPort = 0<br>ResetLimitCounter = 1 (reload rate)<br>JoinReqReloadRate = 127<br>NotifyReloadRate = 127<br>GlobalUplinkReloadRate = 127<br>OverallReloadRate = 127<br>FwdLimitLoadCapacityPL = 0xFF<br>Payload = [0x]45XXXXXXXXXX |  |
| 25 | Relay must send an Unconfirmed frame to the TCL                       |   | → |  | MAC-CMD ConfigureFwdLimitAns<br>Payload = [0x]45<br><br>Note: This response may be piggy backed with the next ED uplink forward                                                                                                                                 |  |
| 26 | ED sends an Unconfirmed frame to the Relay                            | → |   |  | PHYPayload                                                                                                                                                                                                                                                      |  |
|    | Relay must send Unconfirmed frame to TCL                              |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXX                                                                                                                                              |  |
| 27 | ED sends an Unconfirmed frame to the Relay                            | → |   |  | PHYPayload                                                                                                                                                                                                                                                      |  |
|    | Relay sends an Unconfirmed frame to the TCL                           |   | → |  | ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXXXXXXX                                                                                                                                              |  |

|    |                                            |   |   |  |                                                                                                                                                                                                           |                             |
|----|--------------------------------------------|---|---|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
|    | The TCL sends Unconfirmed frames           |   | ← |  | PHYPayload includes<br>Confirmed downlink<br>CP-CMD RegionalDutyCycleCtrlReq-ON<br>FPort = 224<br>Payload = [0x]0501<br><br>ForwardDownlinkReq (PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXX | Turn ON Regional Duty Cycle |
| 28 | ED sends an Unconfirmed frame to the Relay | → |   |  | <i>PHYPayload includes</i><br>ACK bit = 1                                                                                                                                                                 |                             |
|    | Relay must send Unconfirmed frame to TCL   |   | → |  | PHYPayload includes<br>ACK bit = 1<br><br>ForwardUplinkReq (UplinkMetadata, Uplink Frequency, PHYPayload)<br>FPort = 226<br>Payload = [0x]XXXXXXXXXXXXX                                                   |                             |

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